

EMTRAIC 2024

SURROGATE MODEL FOR PEDESTAL PRESSURE

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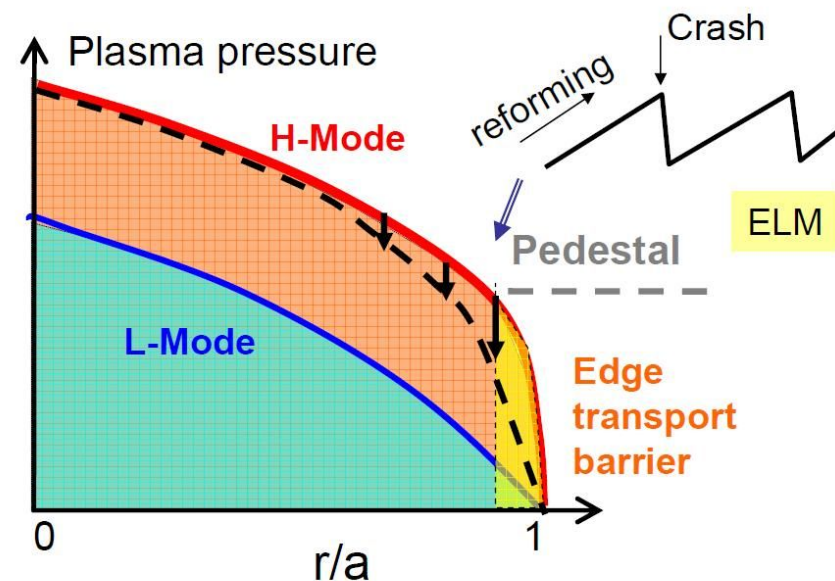
Supervisors: M.SOS,M. KOMM
Special thanks to: J. Cavalier

ELMs? What are they?

Edge Localised Modes (ELMs) are quasi periodic outbursts of hot and dense plasma towards the plasma-facing components (PFCs)

When do they occur?

- in High Confinement Mode (H-mode)
- near the Last Closed Flux Surface (LCFS)
- due to high density and temperature gradients

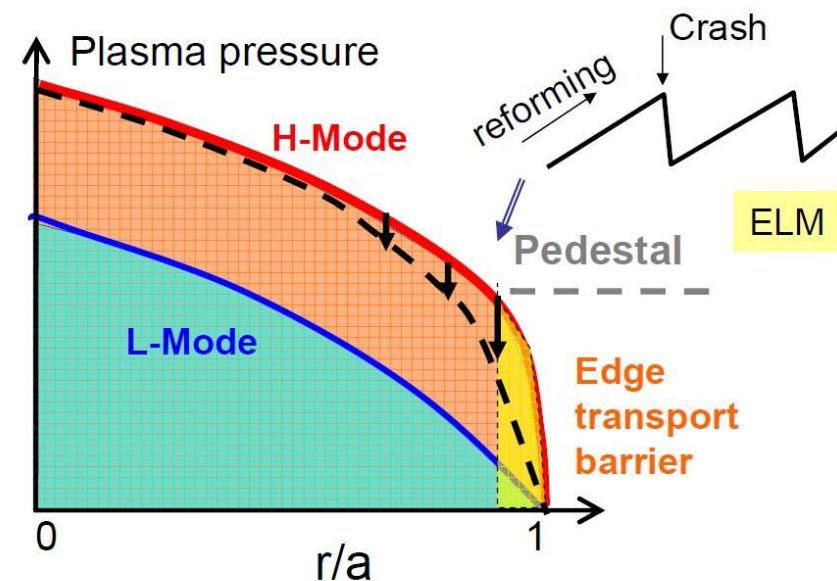


ELMs? What are they?

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Why study ELMs?

Impact of ELMs can damage PFCs
→ risk for future reactors

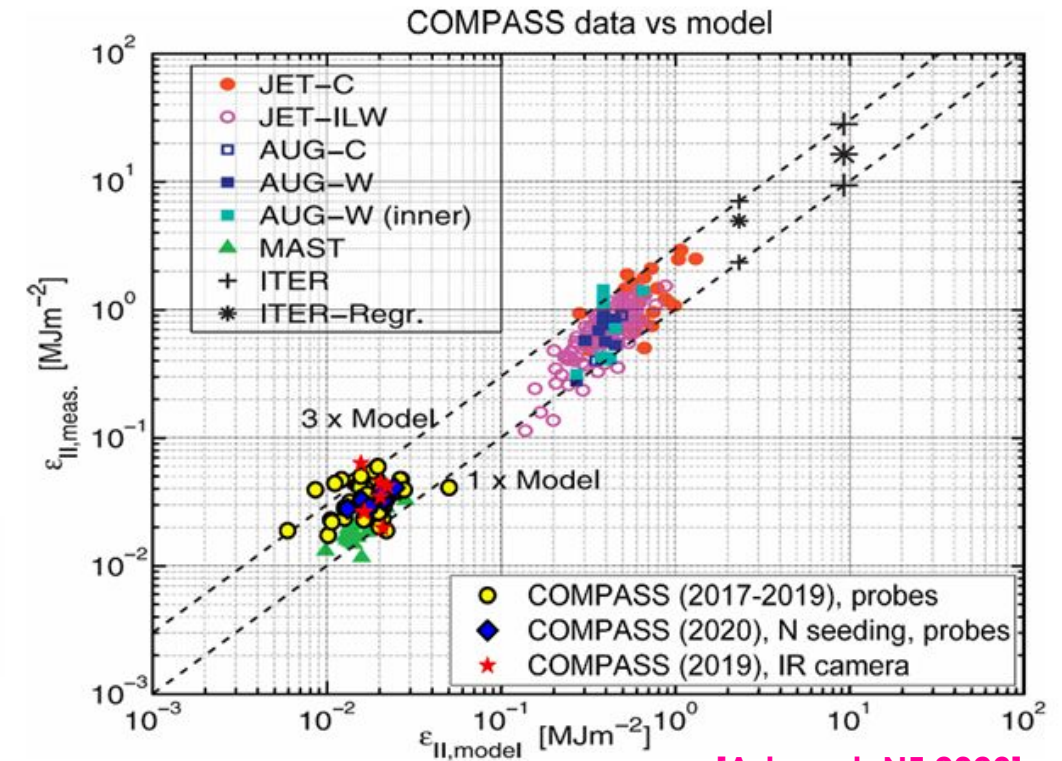


Peak ELM Energy Fluence

maximum value of energy fluence deposited at the divertor during ELM

A predictive scaling for ϵ was proposed by **[Eich NME 2017]**:

$$\epsilon_{II} = \Delta_{equi} \cdot 2\pi a_{geo} \sqrt{\frac{1 + \kappa^2}{2}} \times \frac{3}{2} \cdot n_{e,ped,top} \cdot k_B \cdot T_{e,ped,top} \times \frac{B_{tor}}{B_{pol}}$$



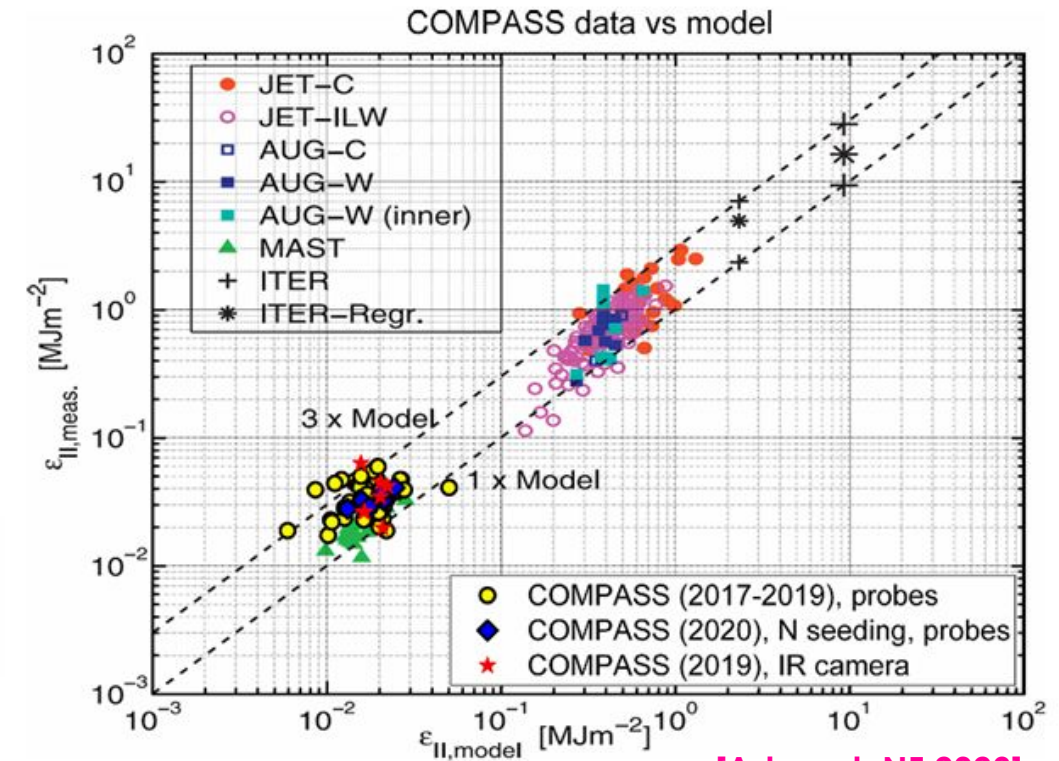
[Adamek NF 2023]

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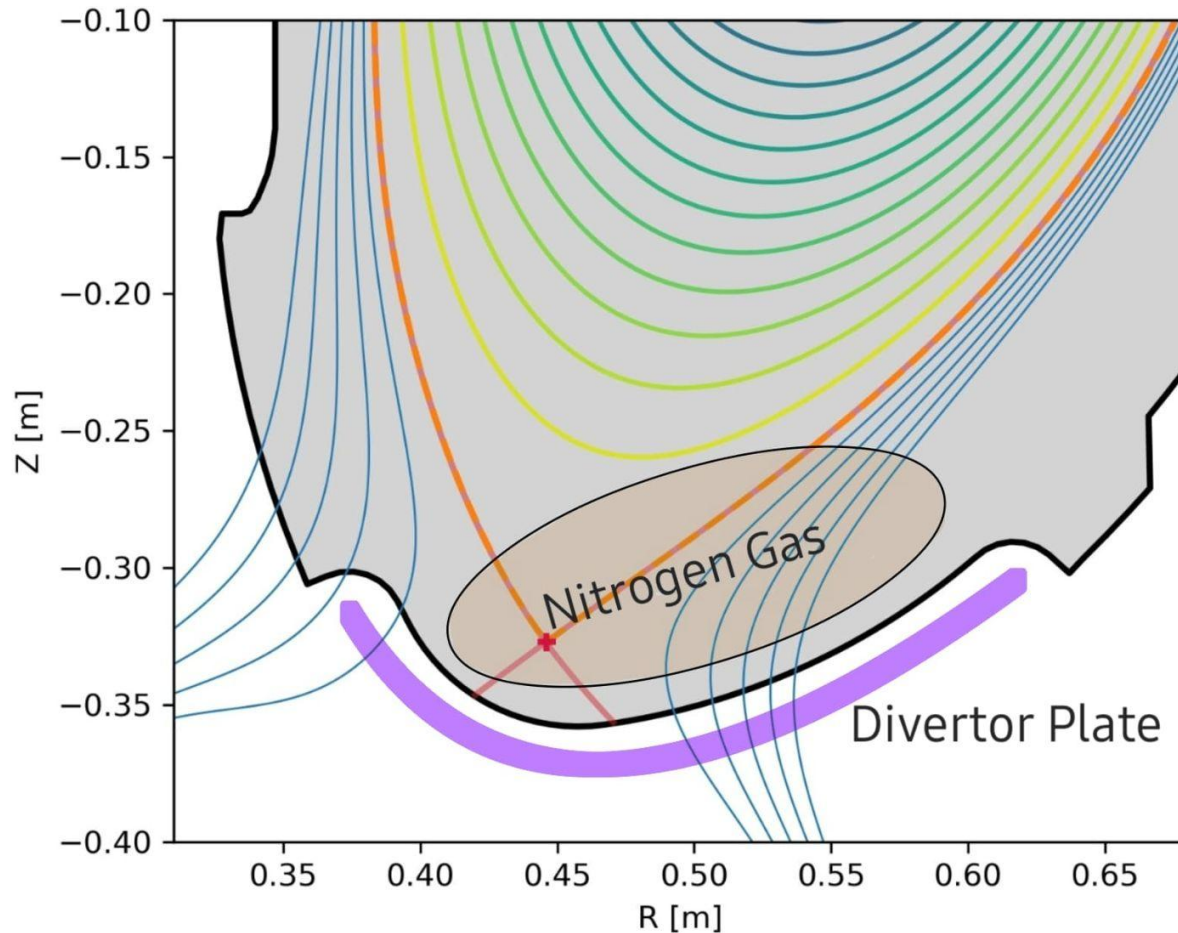
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[Adamek NF 2023]

The predictions for ITER are pessimistic: at least 3x above the material limit.



How to reduce $\epsilon_{||}$?

Mitigation by injection of impurities which have favourable radiative characteristics

How to demonstrate that?

$\epsilon_{||}$ vs time



with N-seeding, $\epsilon_{||}$ must reduce.

Eich Scaling

A predictive scaling for ε was proposed by [Eich NME 2017]:

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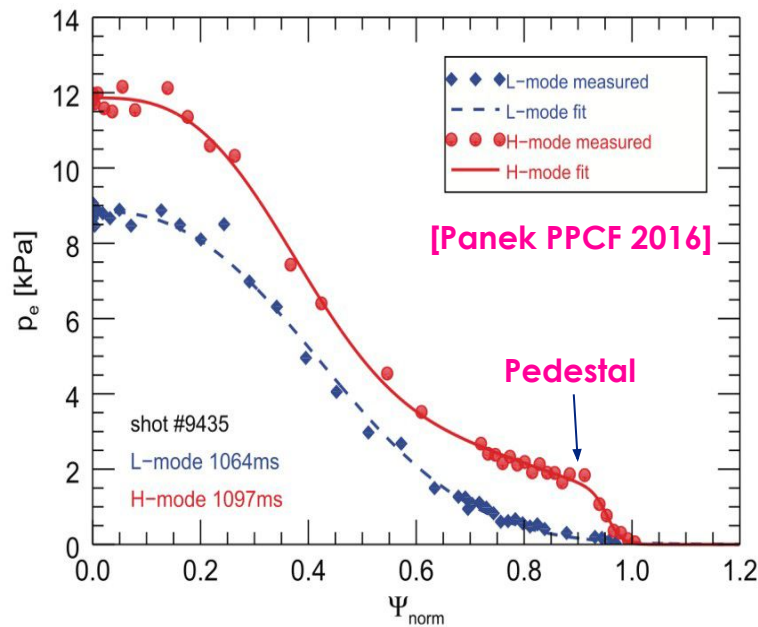
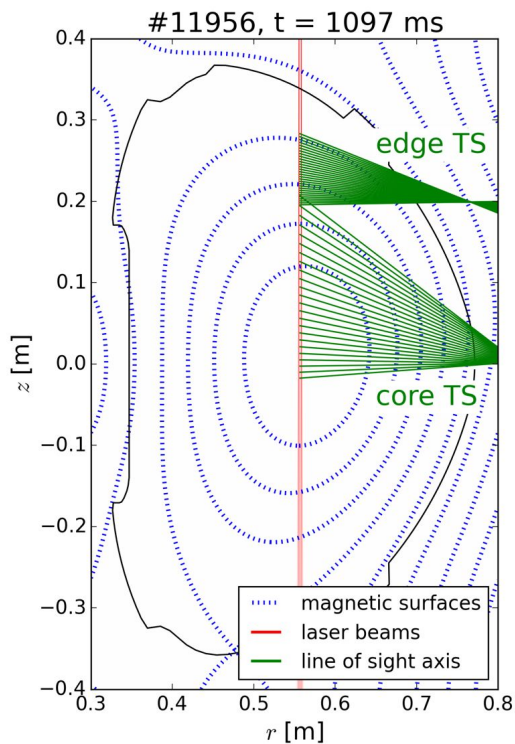
Pedestal Pressure

Elastic scattering

free electrons



photons

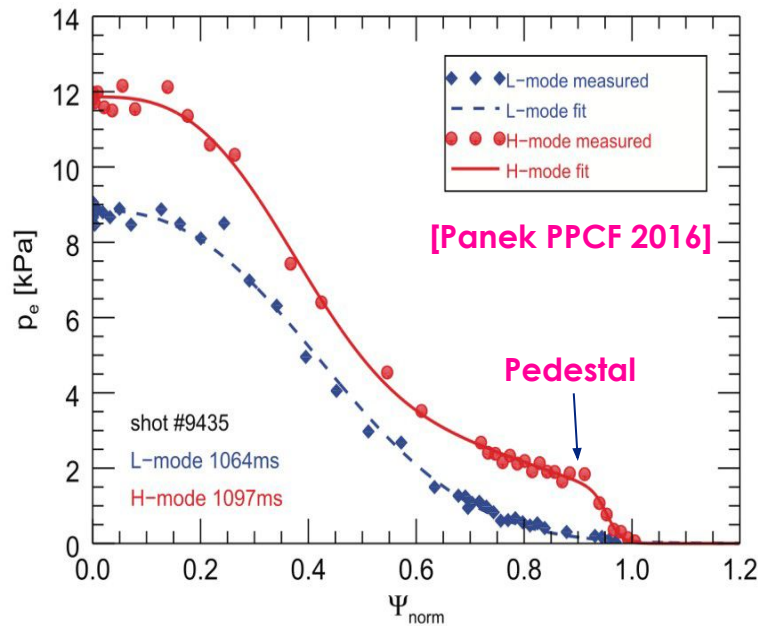
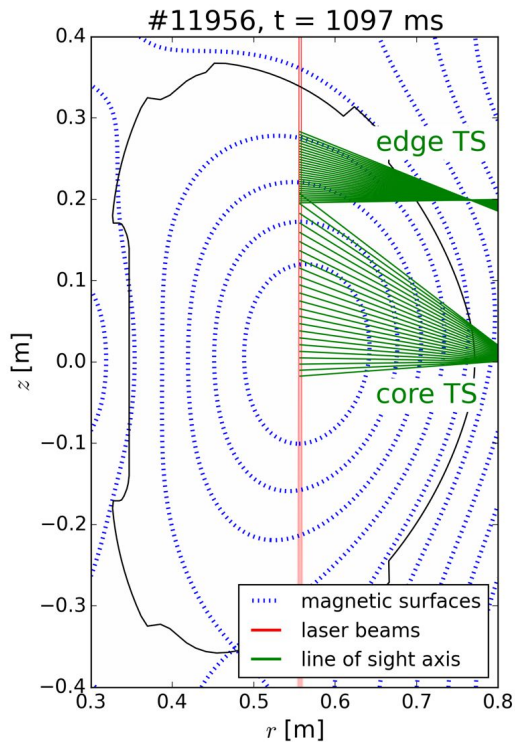


Elastic scattering

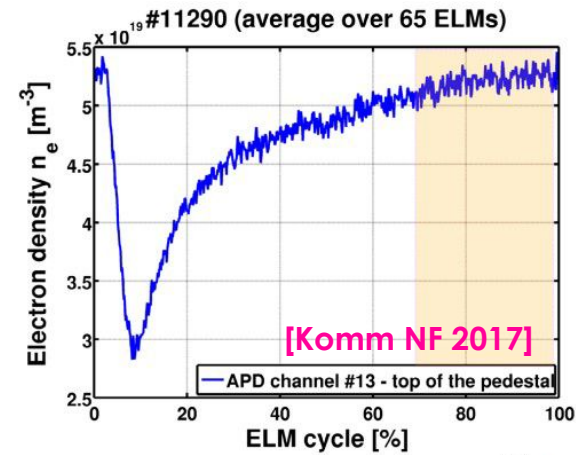
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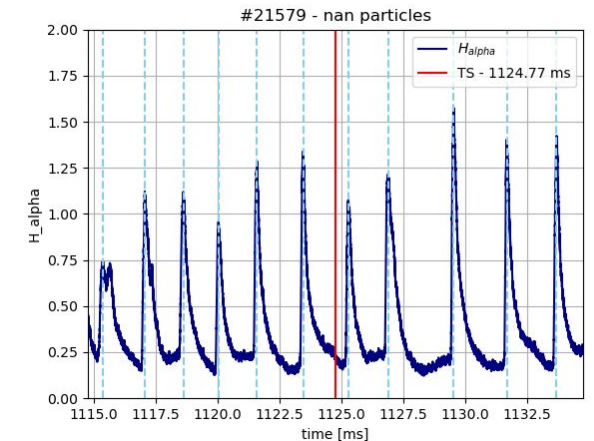
Low temporal resolution !



Repetition rates ~ 10ms

Last 30% of ELM cycle

1-2 measurements

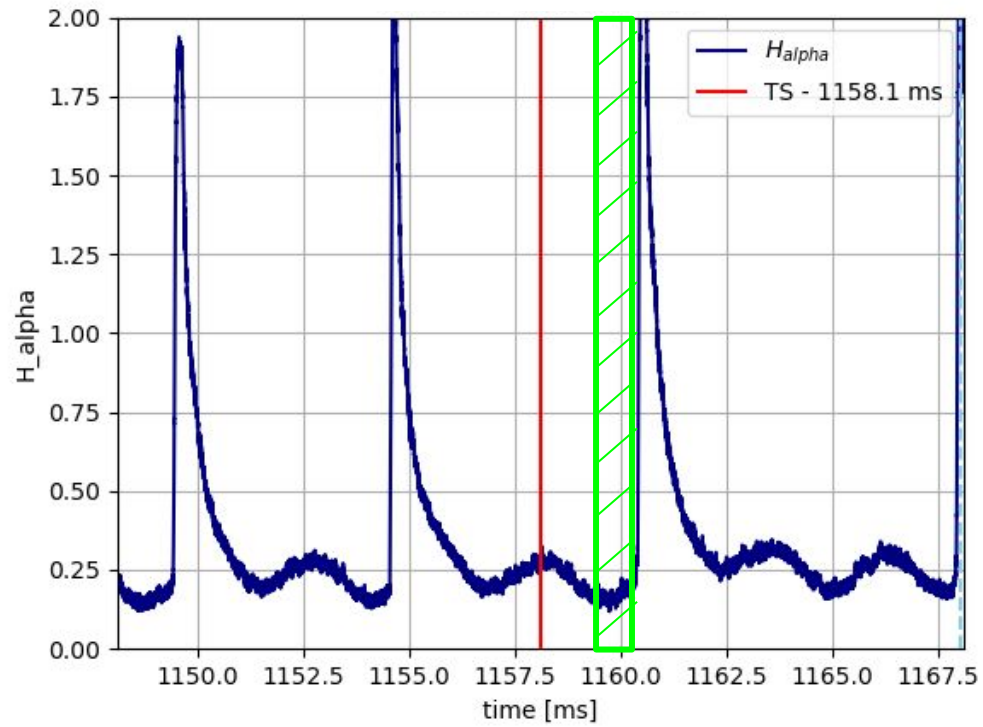






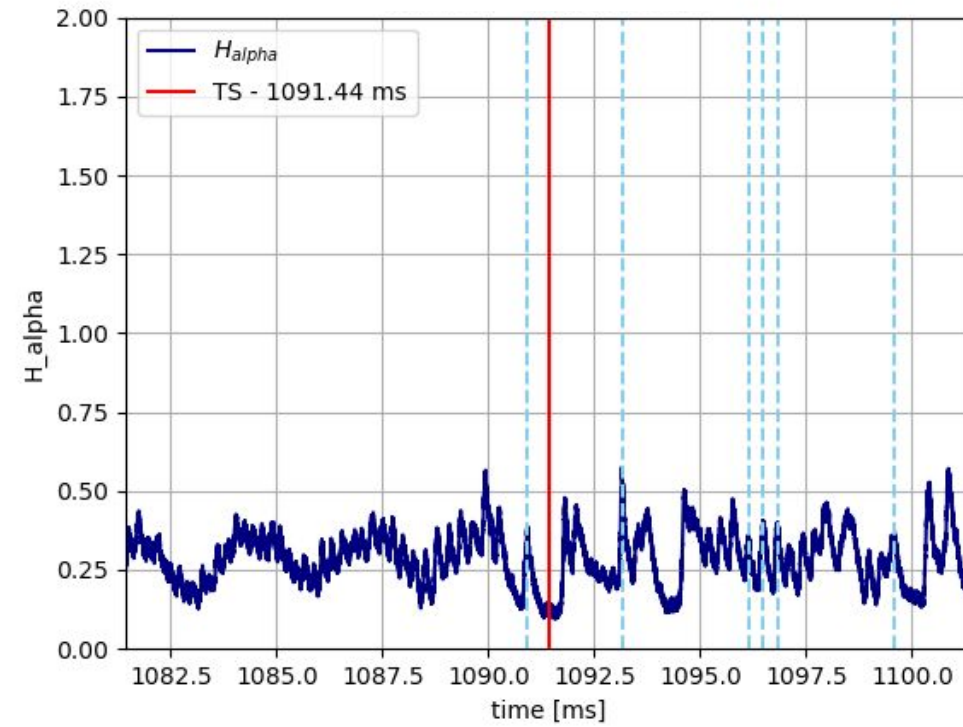


#21588



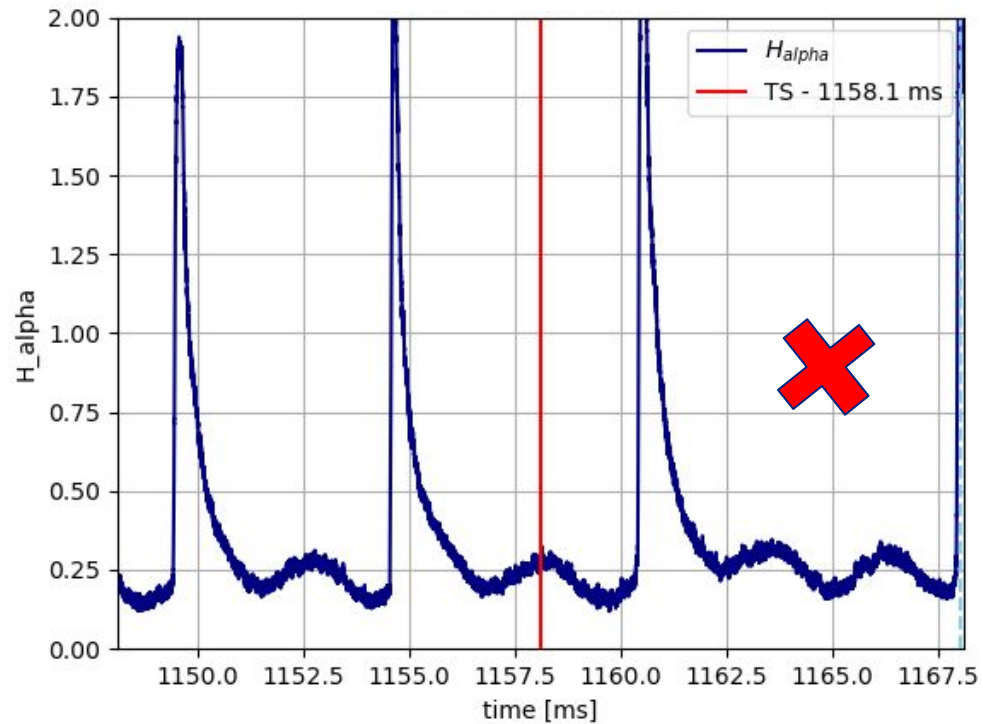
Wrong automatic detection

#21807



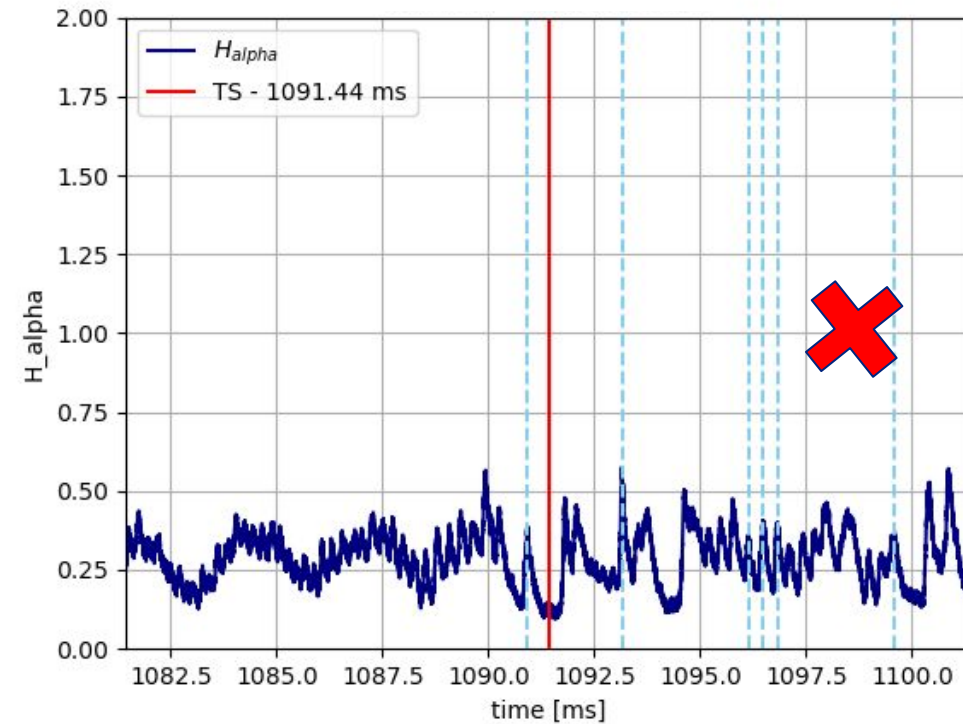
Not ELM signal

#21588

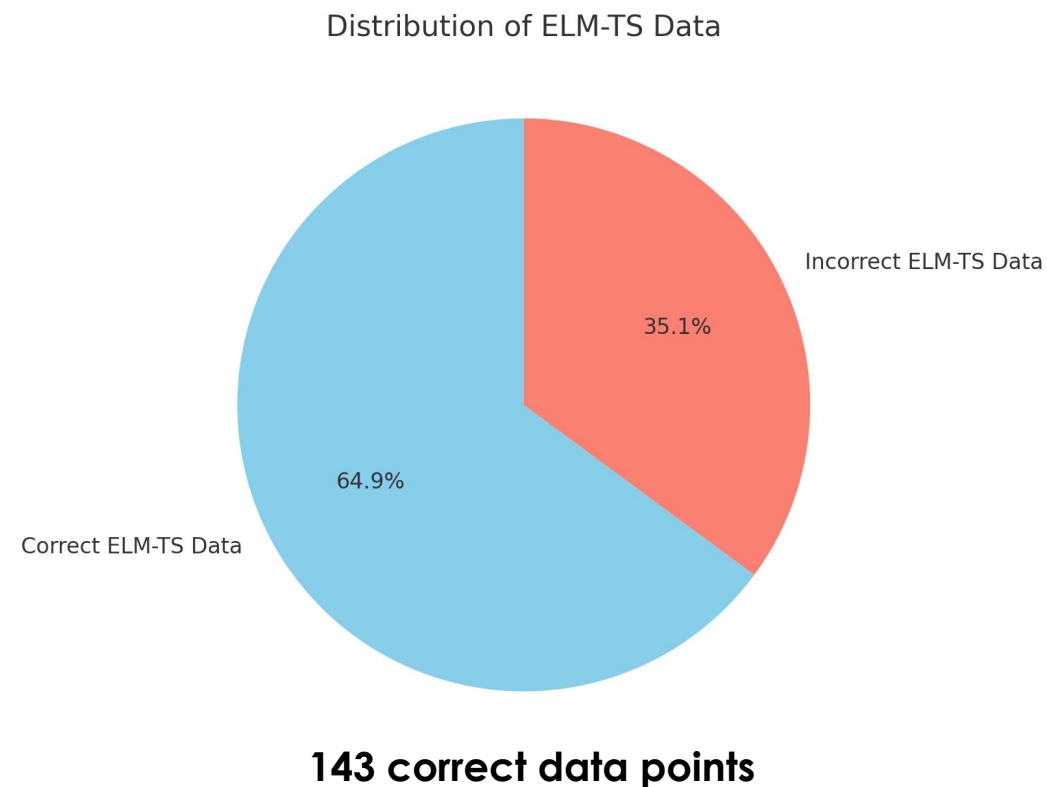
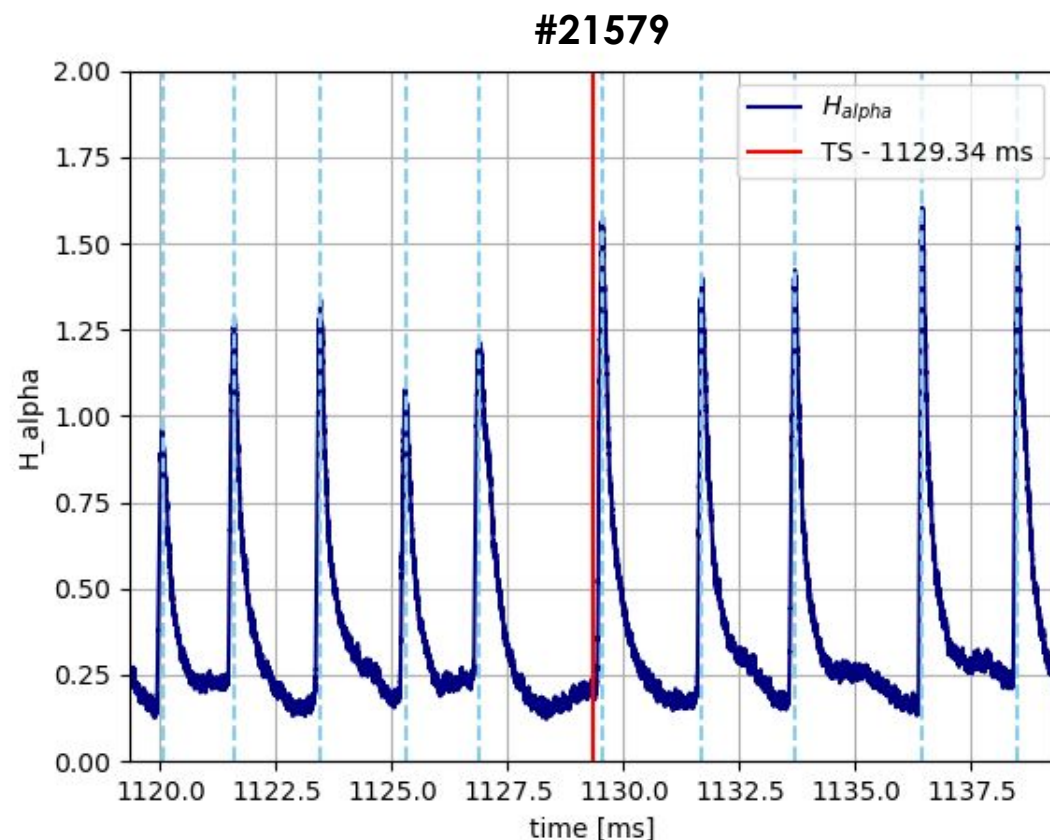


Wrong TS detection time

#21807



Not elm signal

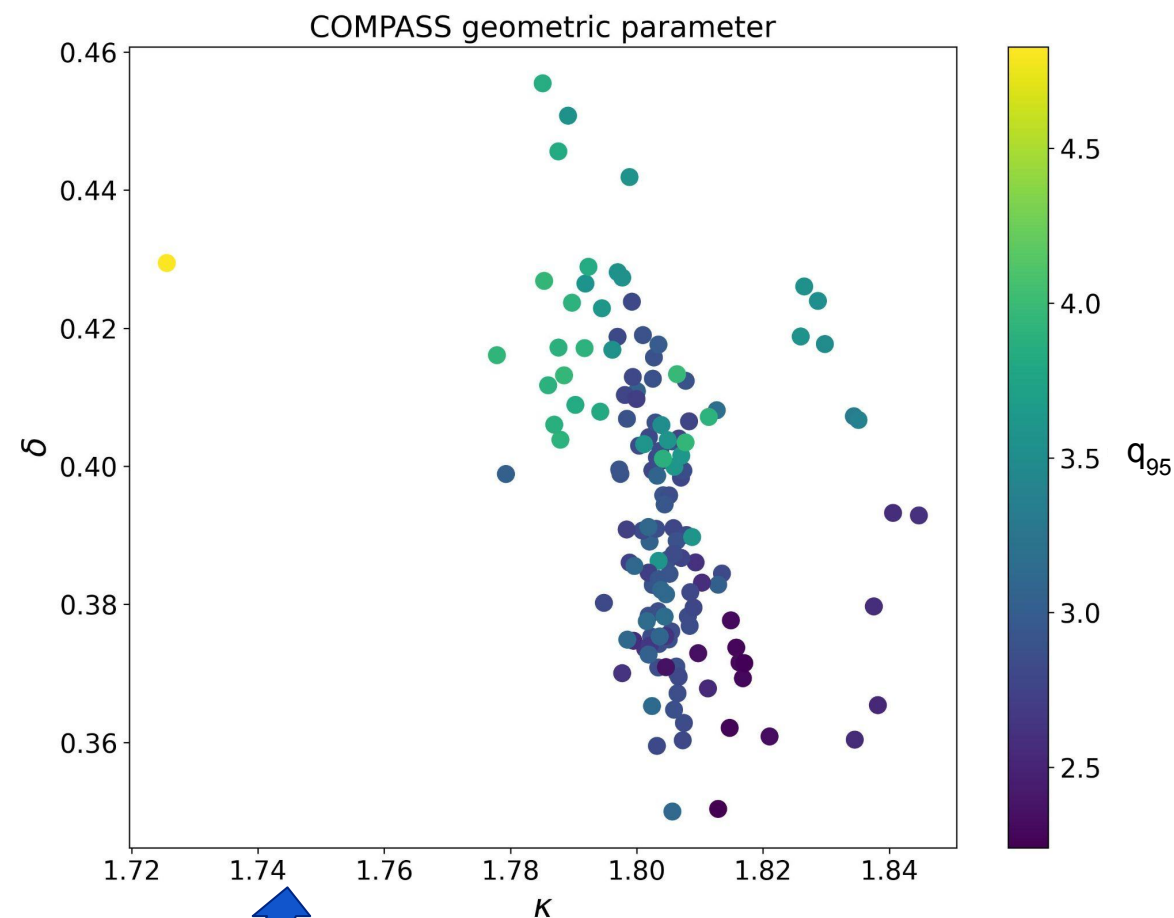


Scaling law:

$$\log P \propto \alpha_1 \log I_p + \alpha_2 B_T + \alpha_3 W_p + \dots$$

15 input signals:

global plasma parameters	$I_p, n_e, W_p, E_{\text{elm}}, \text{beta}, P_{\text{ohm}}, f_{\text{elm}}, R, B, H_{\text{alpha}}$
operating parameters	$T_{\text{boronization}}, P_{\text{NBI}}$
geometric parameters	$q_{95}, \text{elongation}, \text{triangularity}$



Pick your favorite model or signal ! ! !

Importance

Most contribution for
pressure?



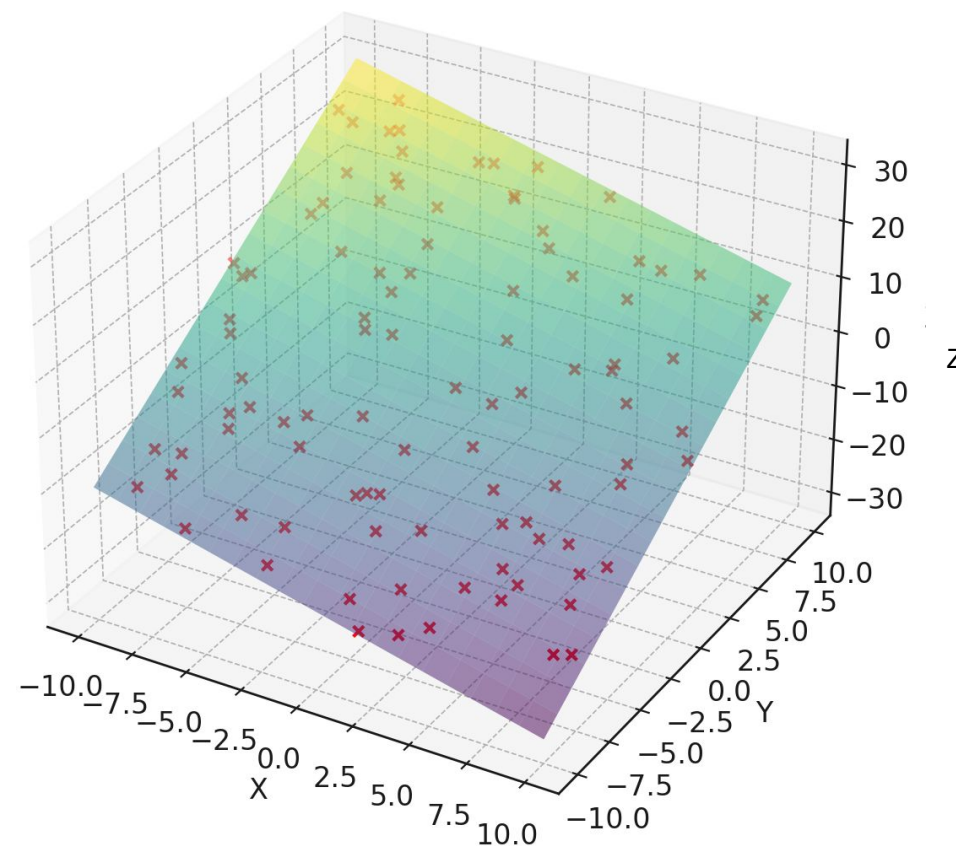
Multicollinearity

Linear correlation between
variables?

Instability of the regression ?

$$Z = -X + 2Y$$

{X is independent to Y}

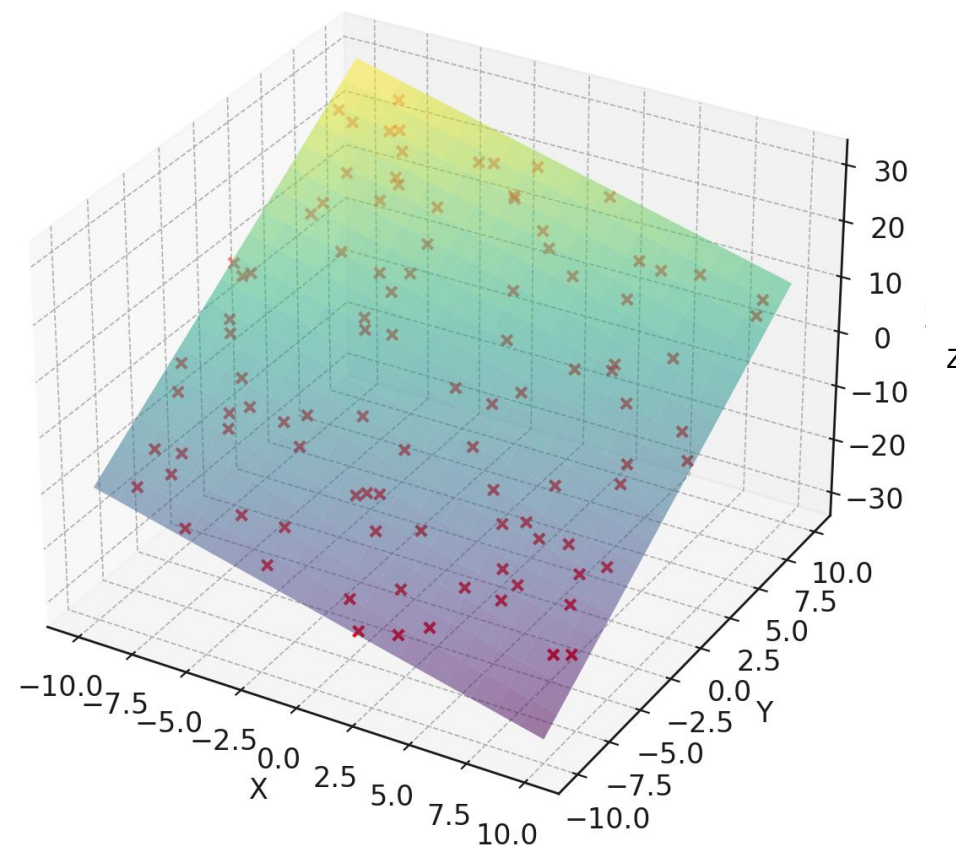


Instability of the regression ?

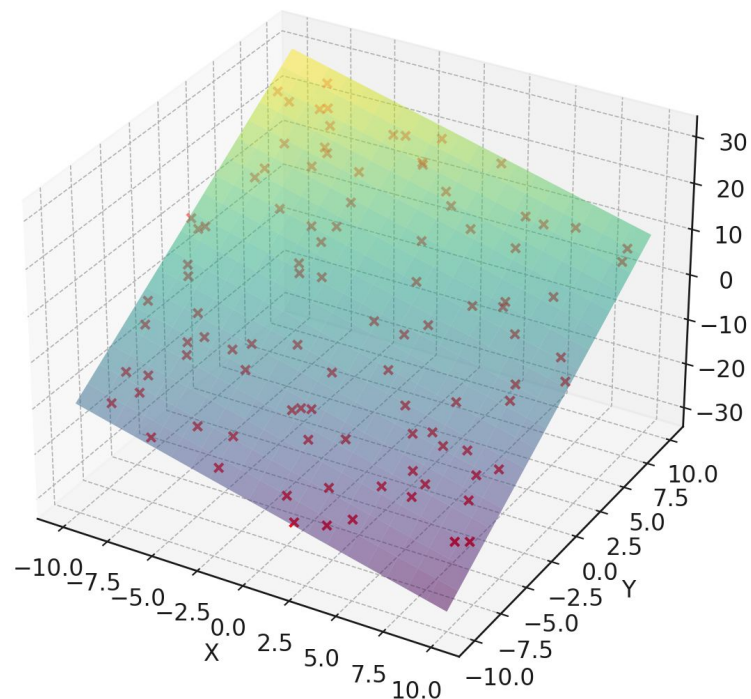
$$Z = -X + 2Y$$

{X is ~~independent~~ to Y}

What's wrong with strong covariance ?

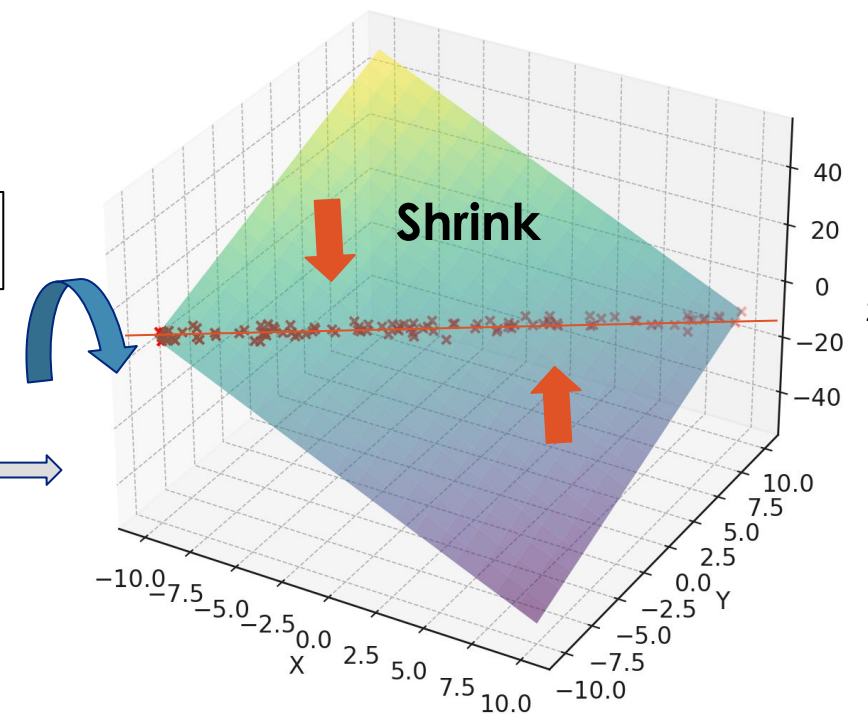


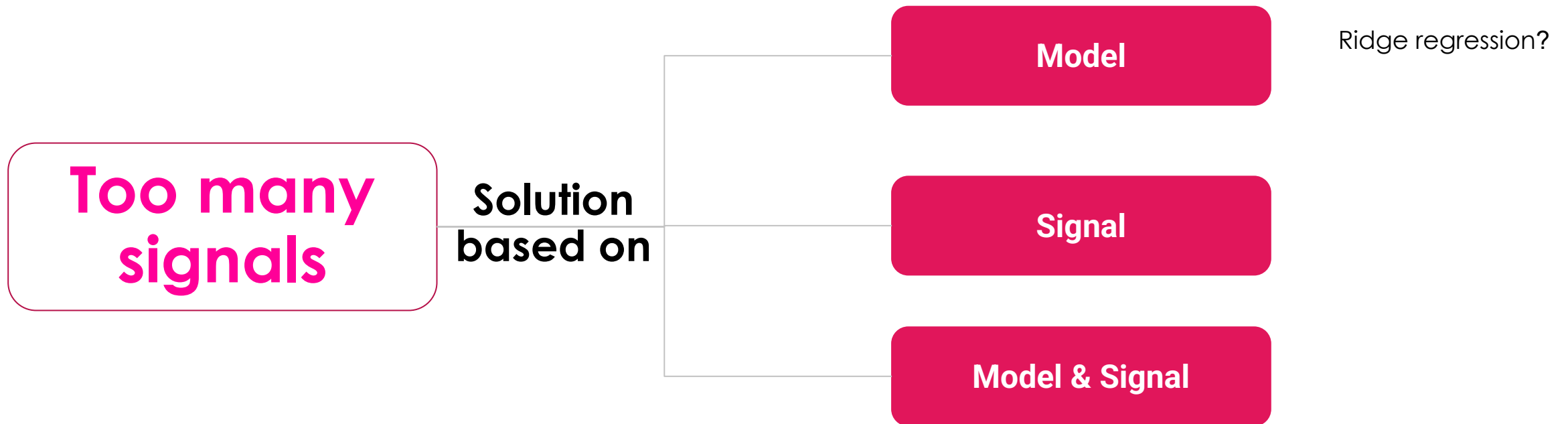
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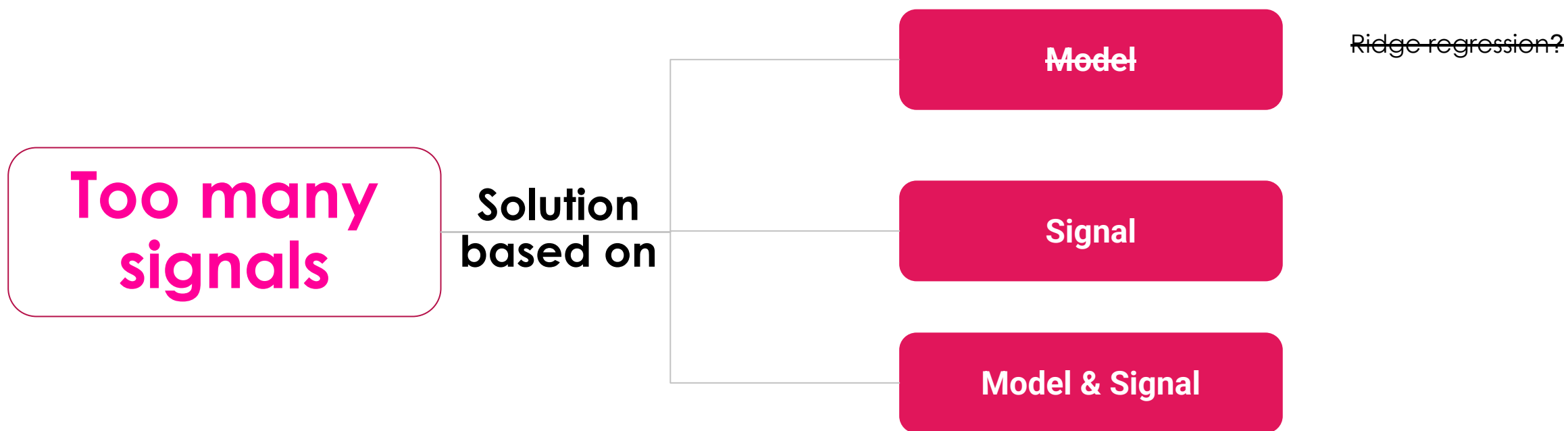


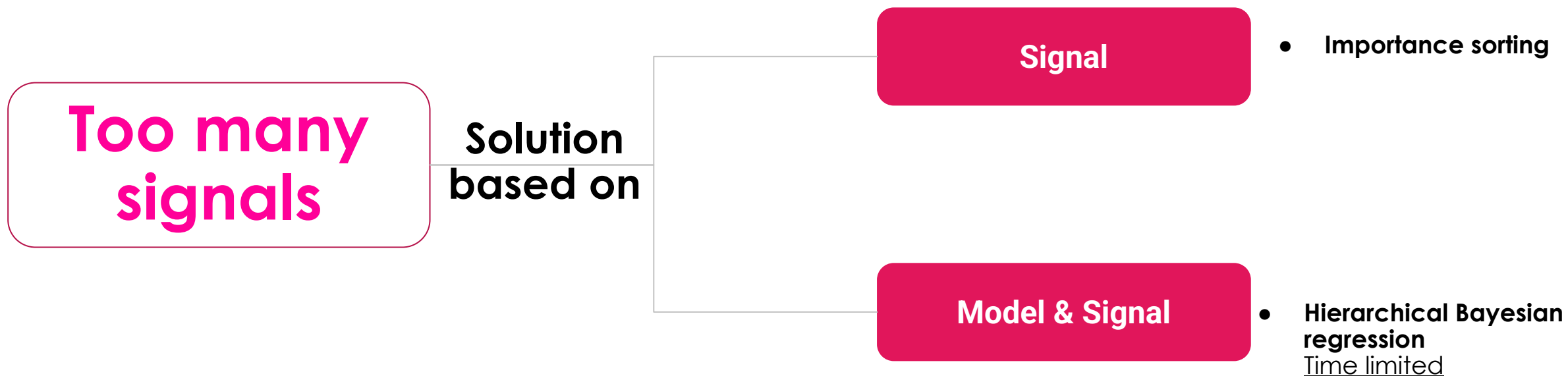
Strong covariance

$$X = 3Y + \sigma, \sigma \sim \mathcal{N}(0, 1)$$





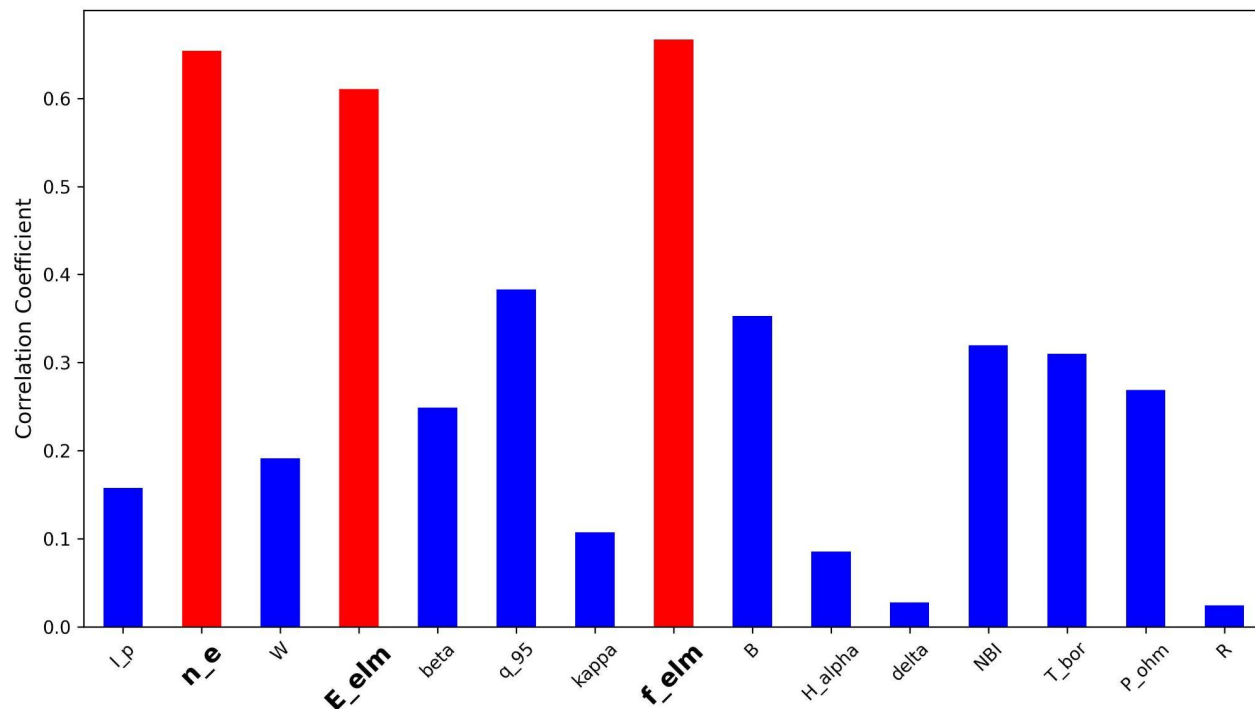




Three Criterias for judging

1: Correlation matrix

Correlation coefficient with pressure



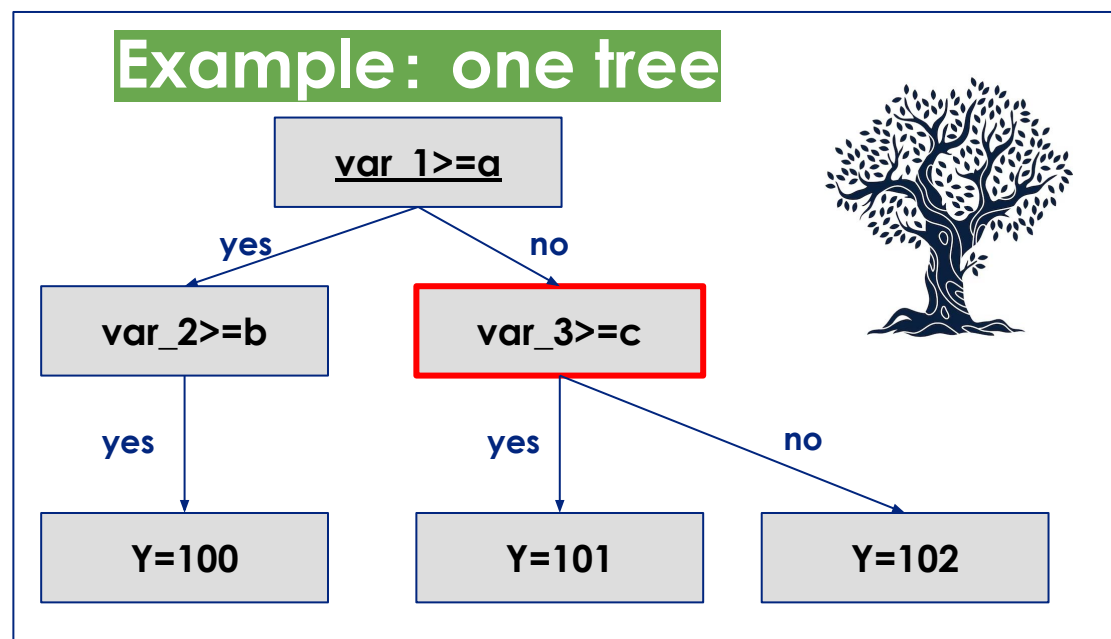
3 top important signals

n_e	line average density
E_elm	Elm energy
f_elm	Elm frequency

Three Criterias for judging

2: Random Forest regression

(Belongs to machine learning)

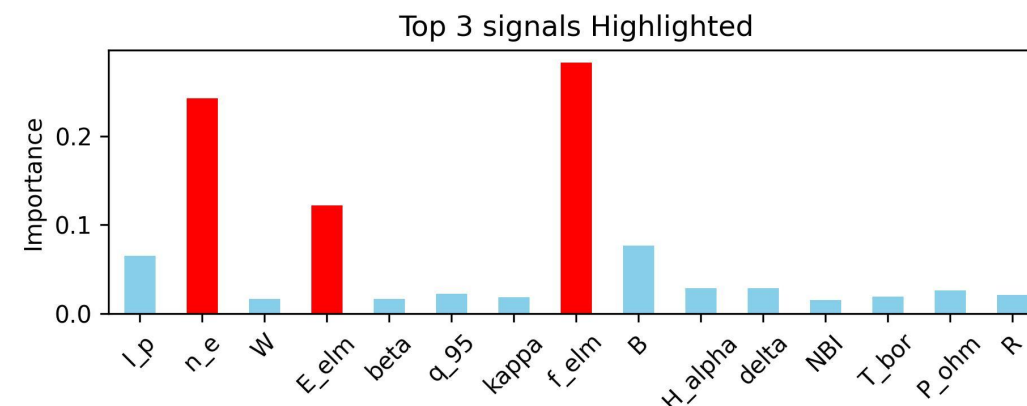
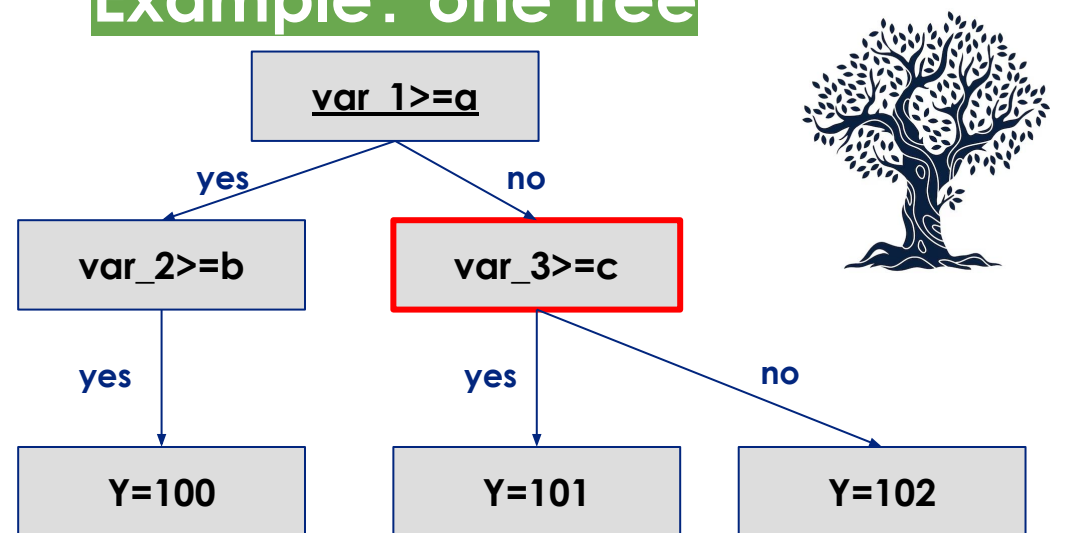


Three Criterias for judging

2: Random Forest regression

(Belongs to machine learning)

Example: one tree



- Importance $\propto$ Number of occurrence

3 Criterias for judging

3: BIC for regression model

$$\text{BIC} = -2 \cdot \ln(L) + k \cdot \ln(n)$$

L=Likelihood



n=Complexities

BIC lower, model better

Example:

Model 1:

Pre	$\propto$	n_e	I_p
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BIC_{without}

Model 2:

Pre	$\propto$	n_e	I_p	B_T
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BIC_{with}



If **Score** = $\text{BIC}_{\text{without}} - \text{BIC}_{\text{with}} > 0$, keep B_T

3 Criterias for judging

3: BIC for regression model

$$\text{BIC} = -2 \cdot \ln(L) + k \cdot \ln(n)$$

L=Likelihood

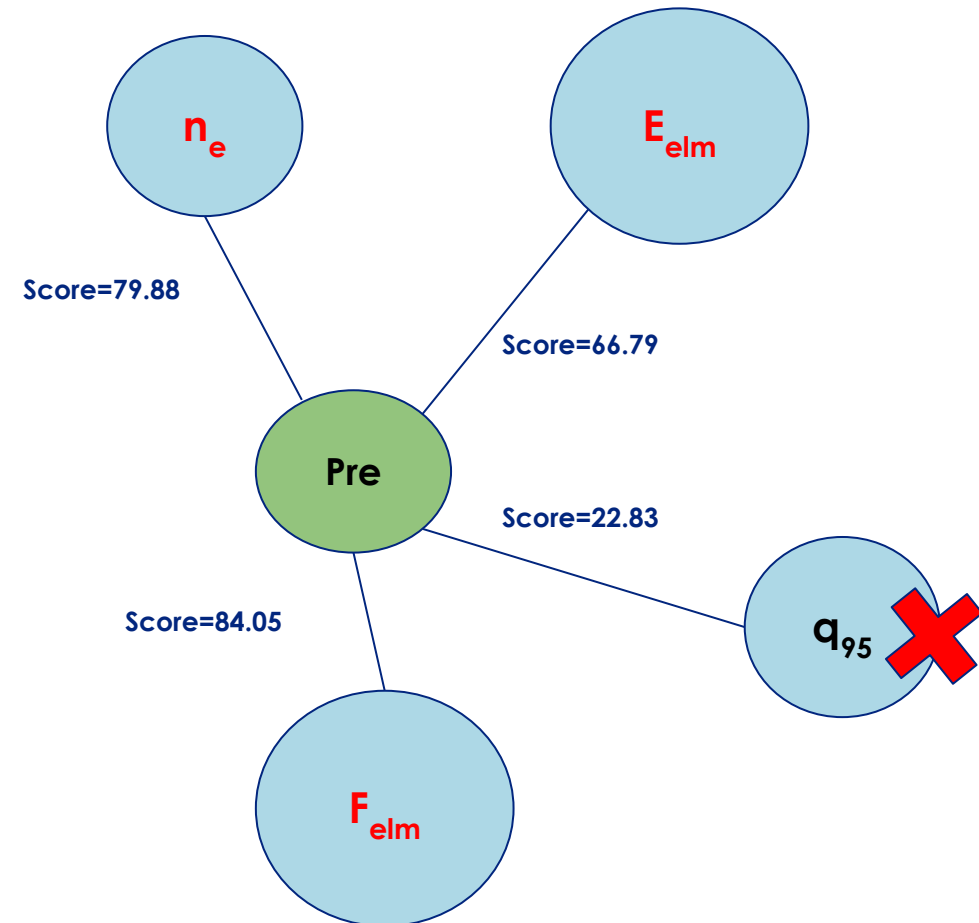


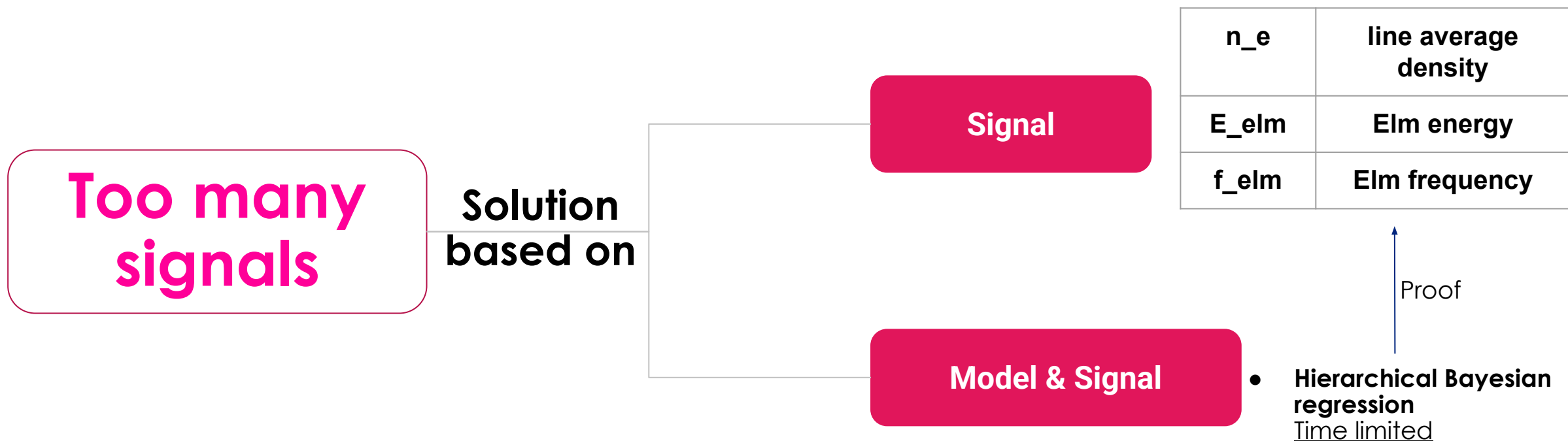
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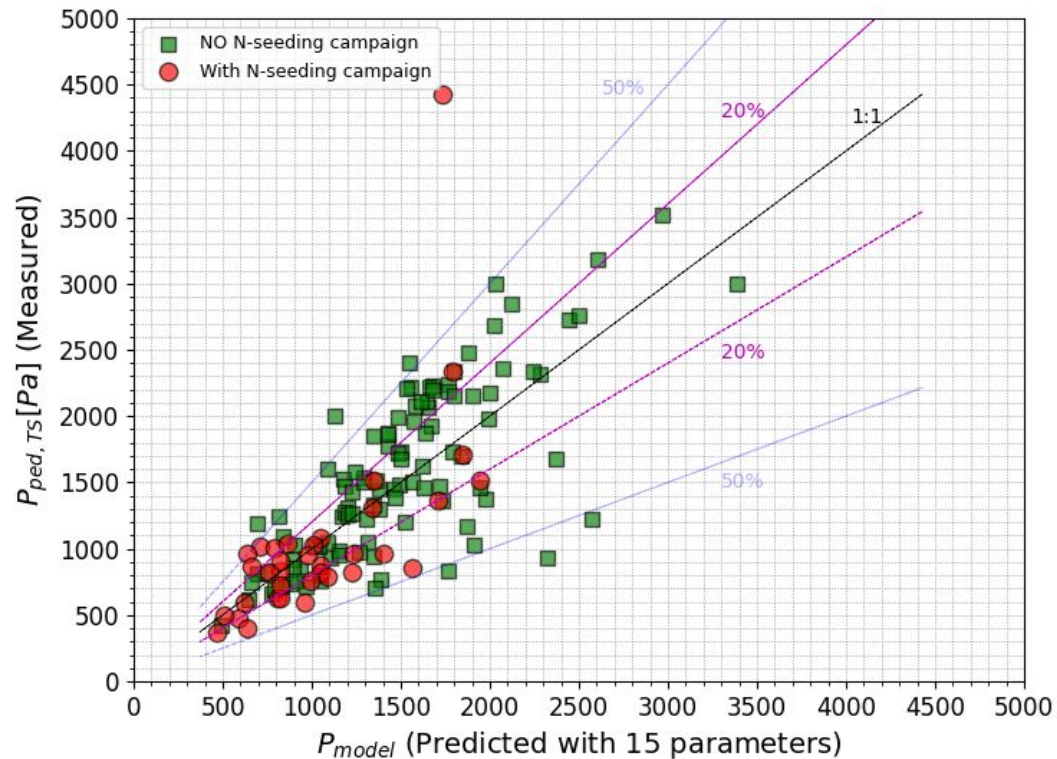
||

Score higher, signal important

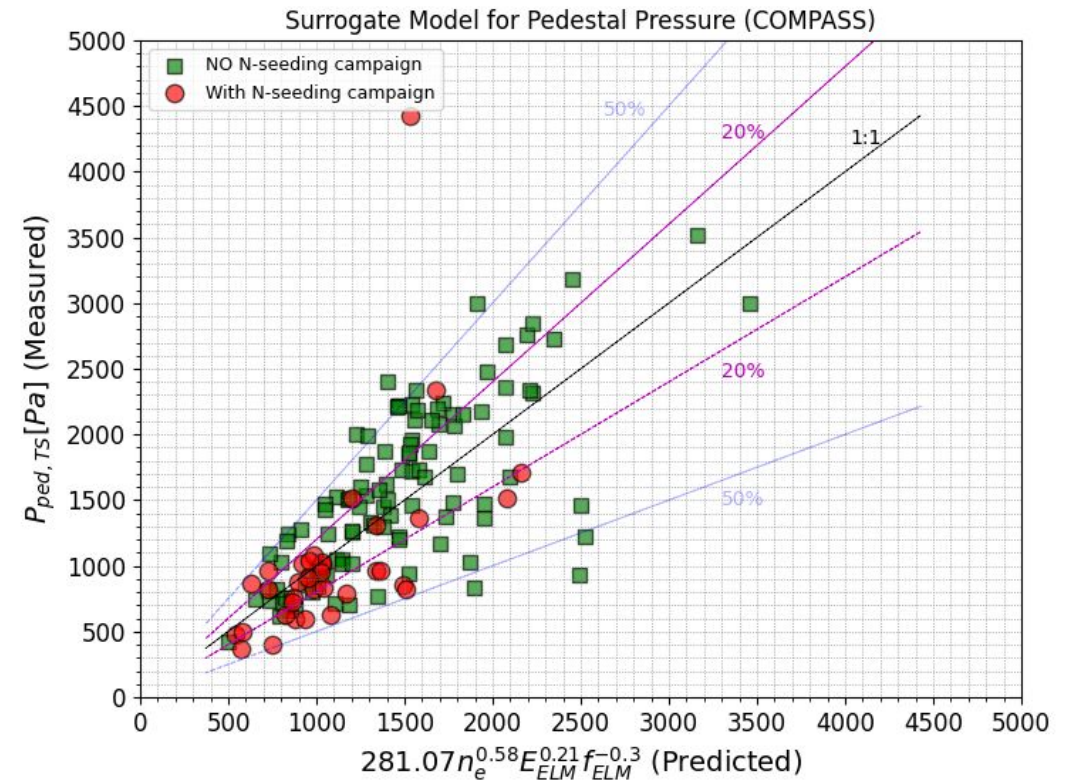




Final validation



R^2 score: 0.655



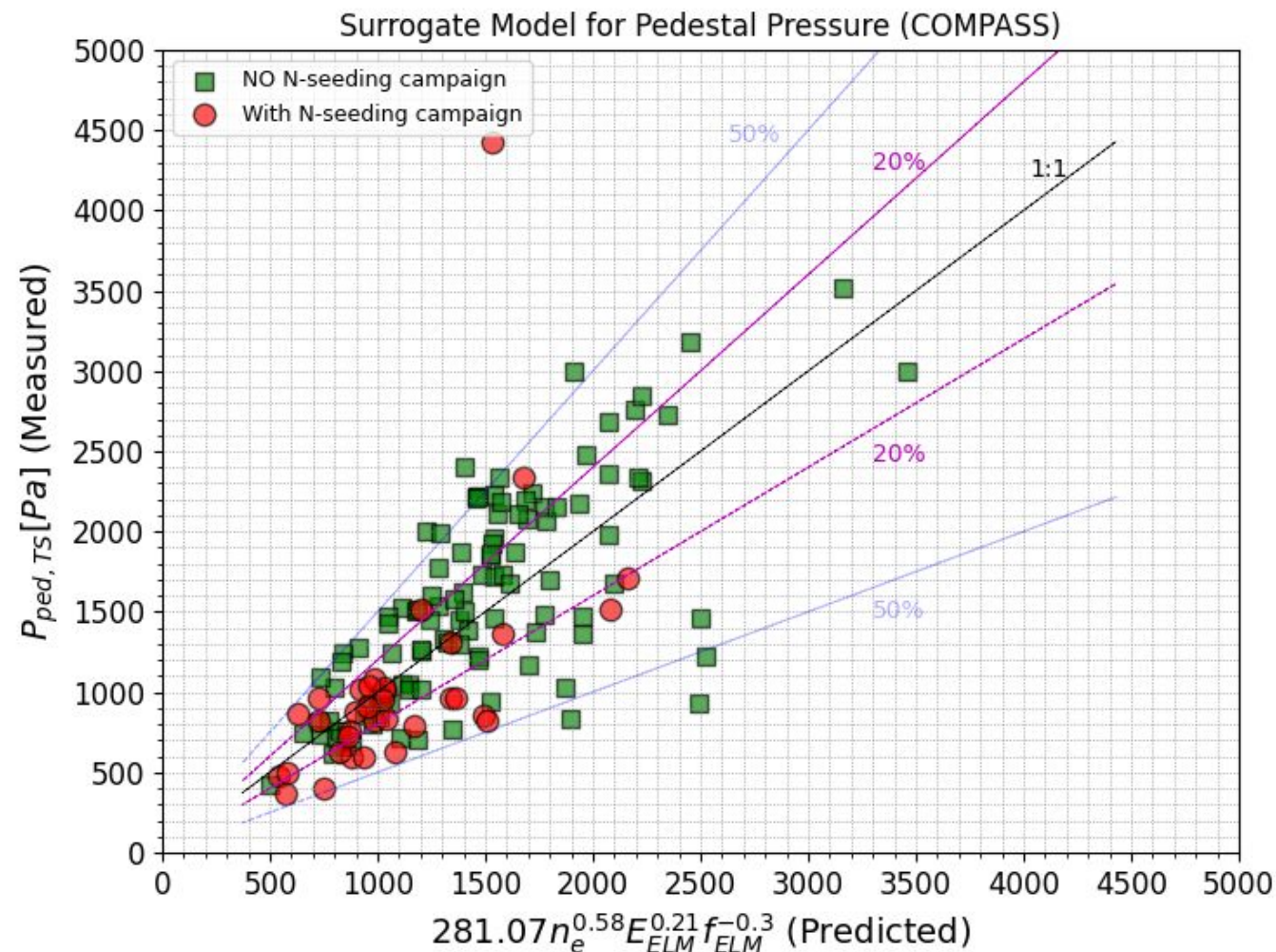
R^2 score: 0.5957

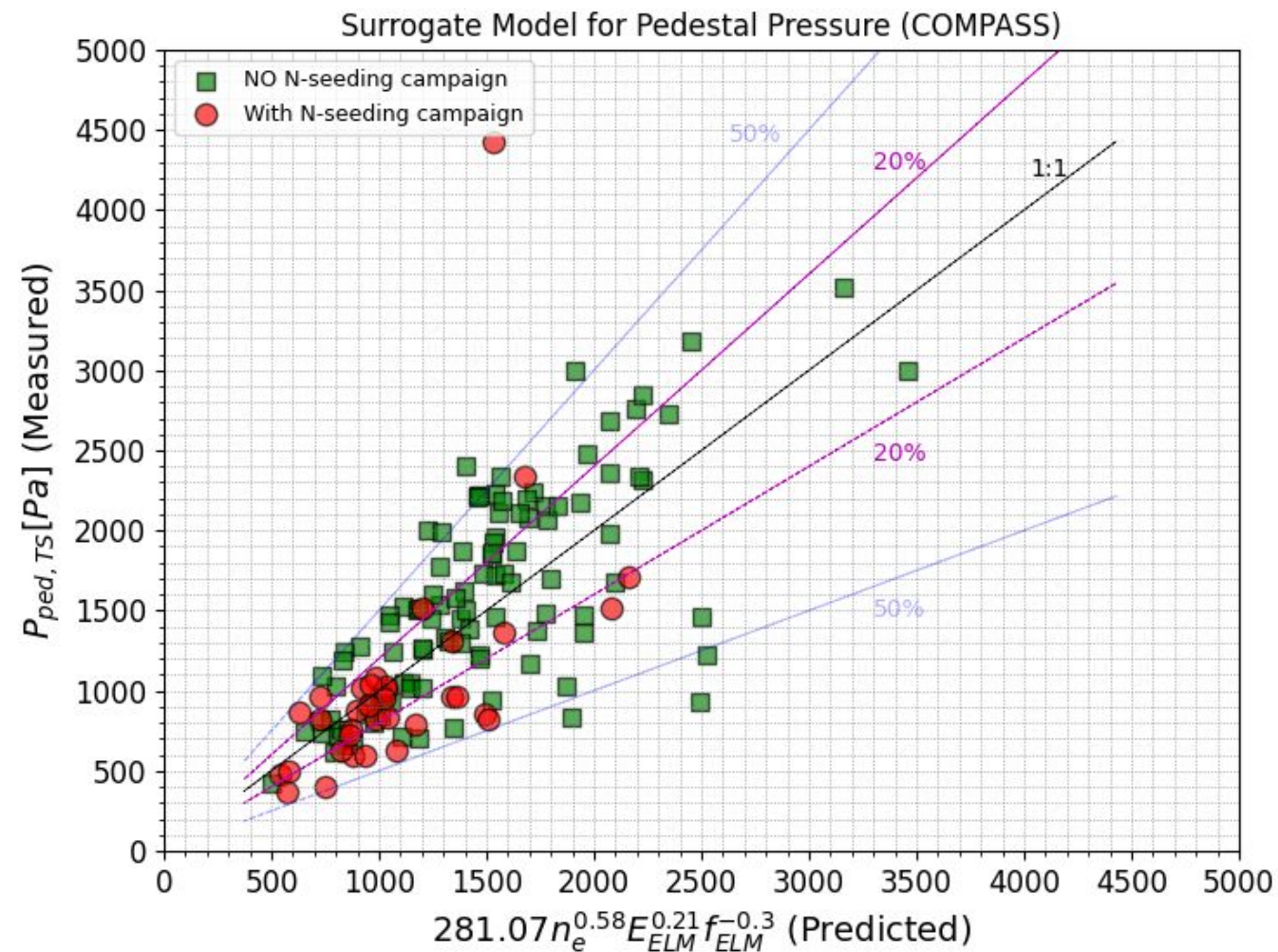
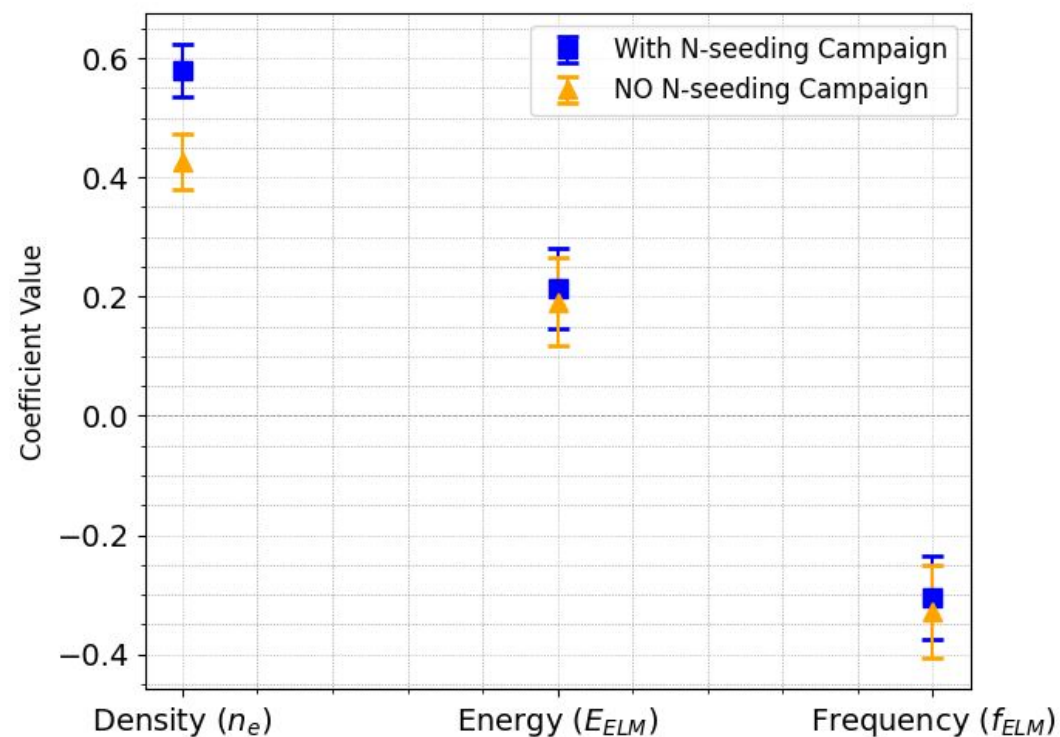
Scaling Law:

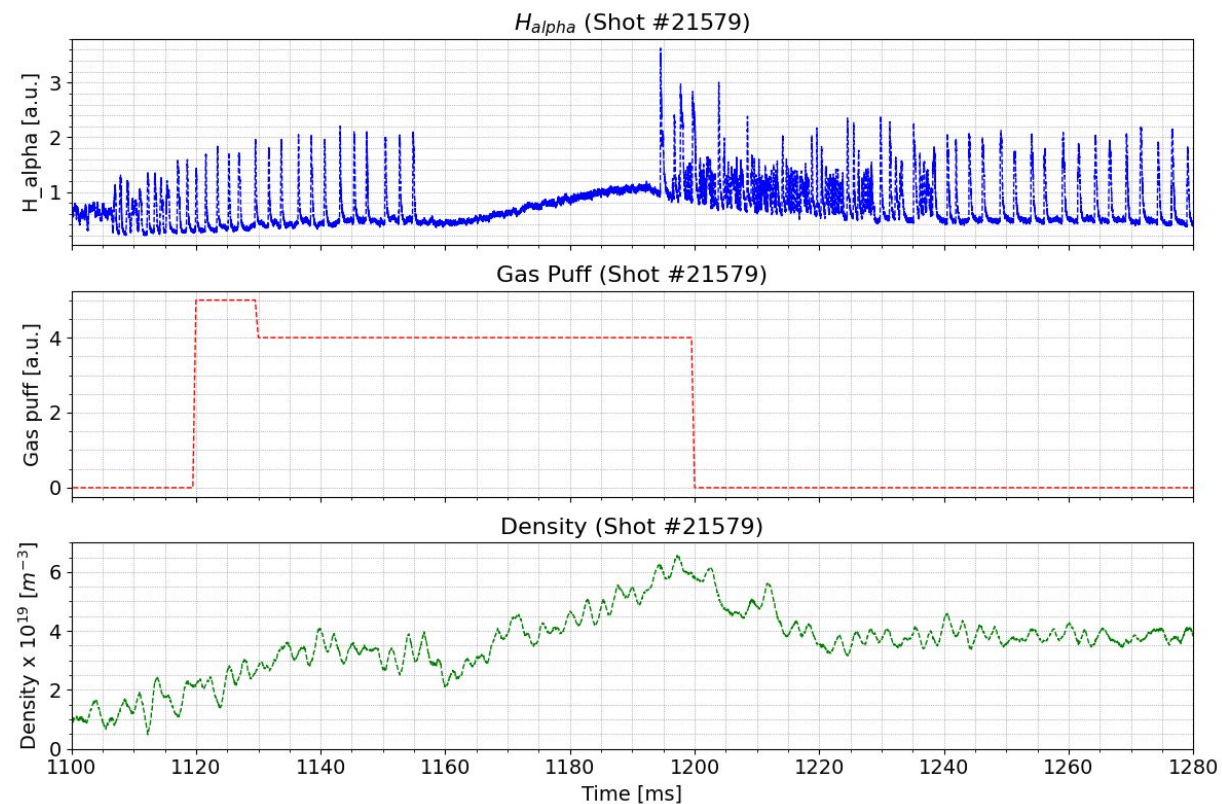
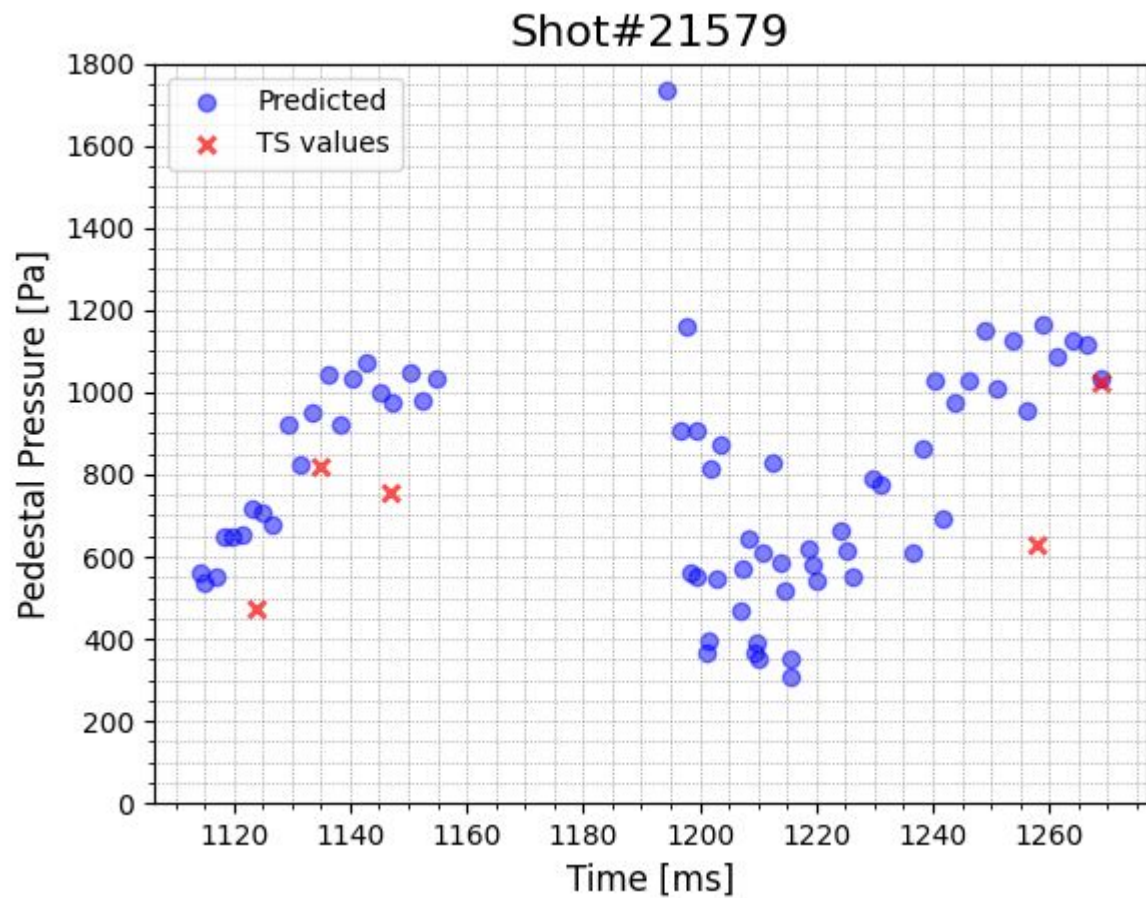
$$P_{ped,TS} = 281.07 n_e^{0.58} E_{ELM}^{0.21} f_{ELM}^{-0.3}$$

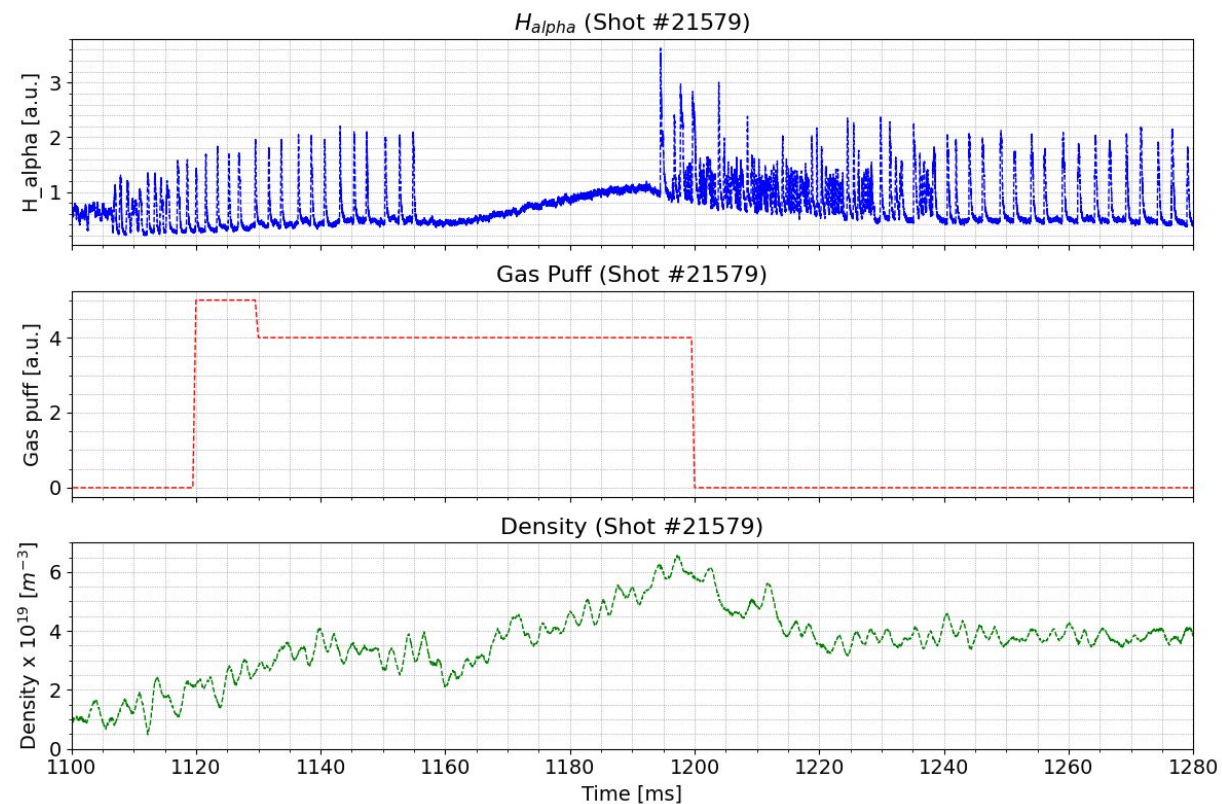
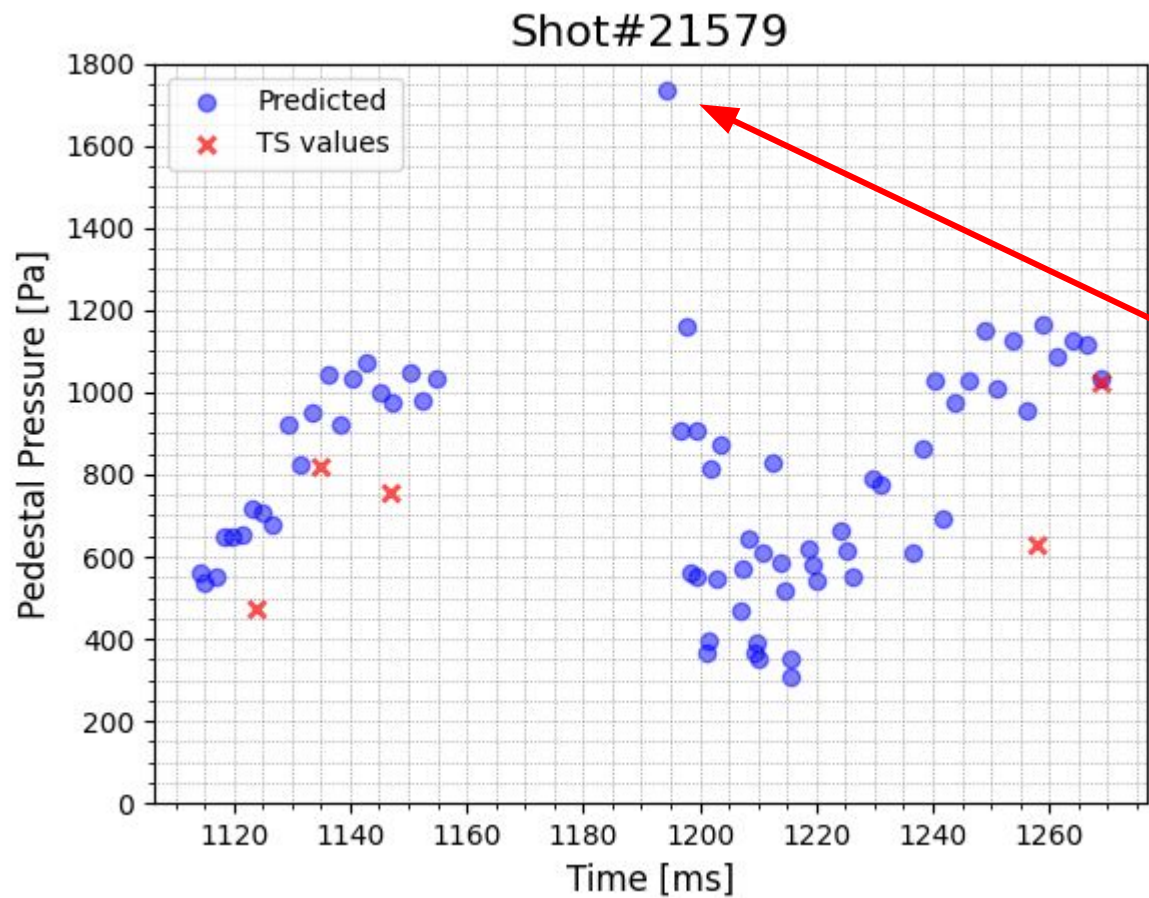
R² score: 0.5957

- Predictions are within 50% error bands.
- Some scatterness for both N-seeded and non-seeded discharges can be seen.

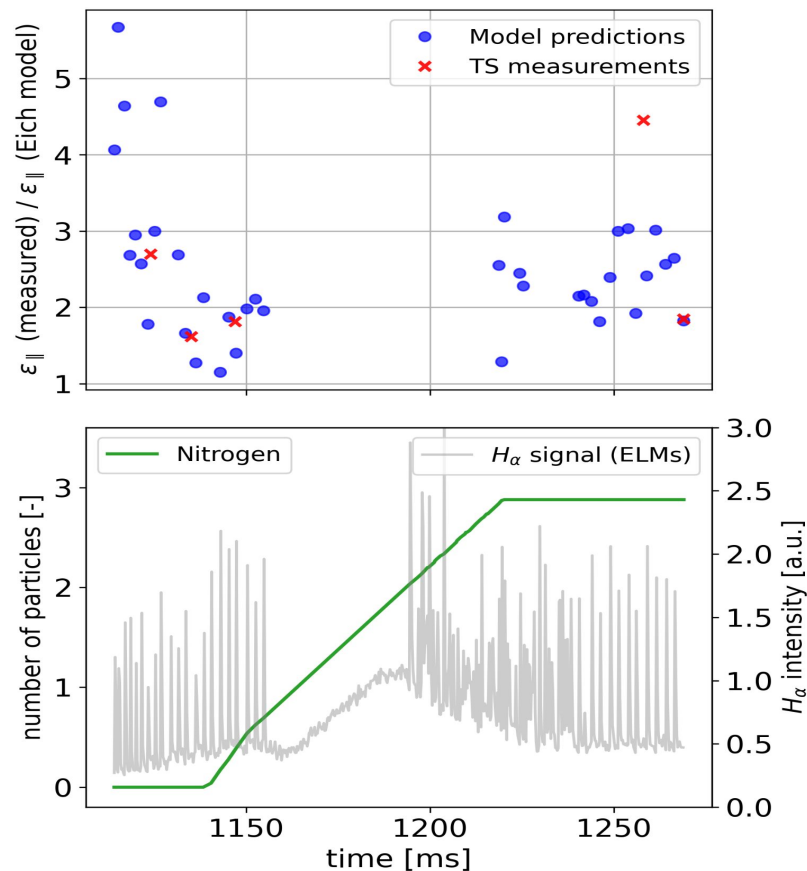








Future work:



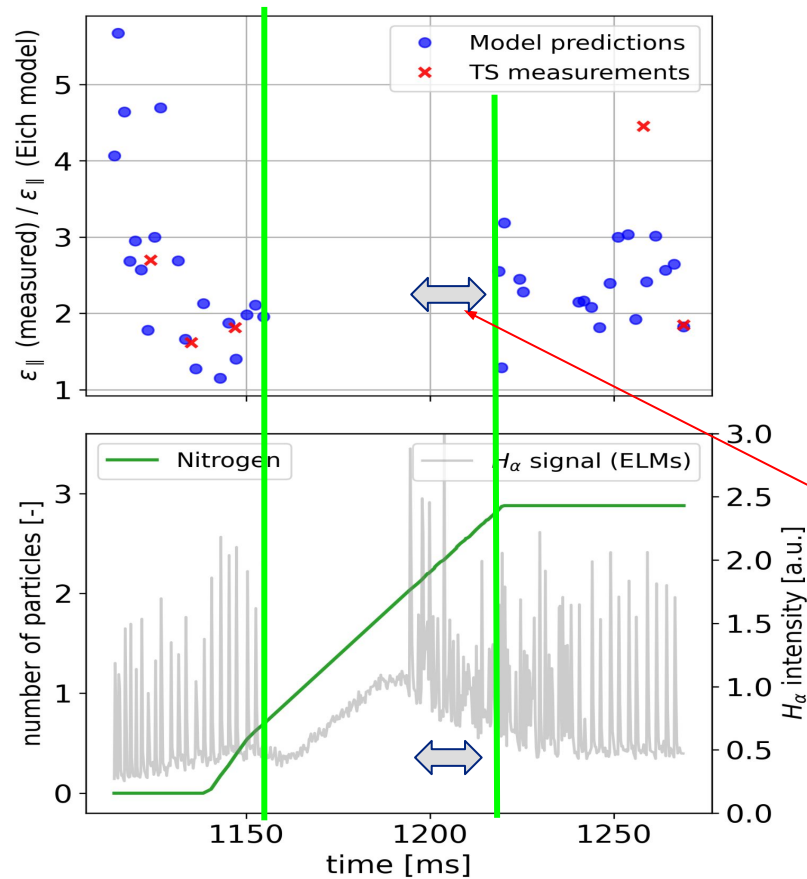
Eich Scaling model:

$$\varepsilon_{||} = \Delta_{equi} \cdot 2\pi a_{geo} \sqrt{\frac{1 + \kappa^2}{2}} \times \frac{3}{2} \cdot n_{e,ped,top} \cdot k_B \cdot T_{e,ped,top} \times \frac{B_{tor}}{B_{pol}}$$

Pedestal Pressure

We Got The high resolution !

Future work:



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$$\varepsilon_{||} = \Delta_{equi} \cdot 2\pi a_{geo} \sqrt{\frac{1 + \kappa^2}{2}} \times \frac{3}{2} \cdot \underbrace{n_{e,ped,top} \cdot k_B \cdot T_{e,ped,top}}_{\text{Pedestal Pressure}} \times \frac{B_{tor}}{B_{pol}}$$

We Got The high resolution !

Lack the probe measured data,
need more work later...

Conclusion

1. Built a 3 parameters scaling law to solve low temporal resolution problem.

$$P_{ped,TS} = 281.07 n_e^{0.58} E_{ELM}^{0.21} f_{ELM}^{-0.3}$$

2. Predict the phase transfer of specific shot

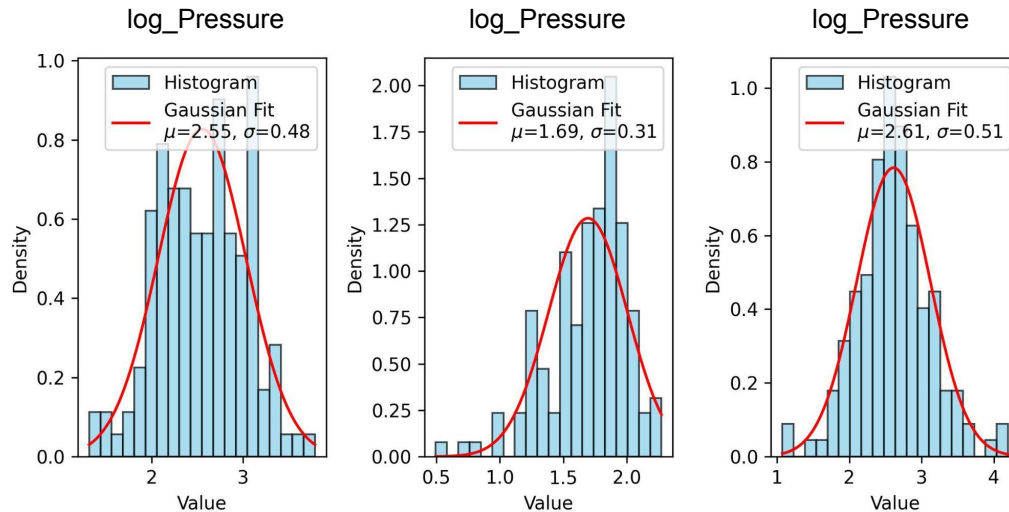
Future Scope

1. More data points for N-seeding can be considered to see the effect of ELM energy reduction.
2. Validity of model can be checked on data from different devices.
3. Cross verification with Machine Learning method — for next talk!



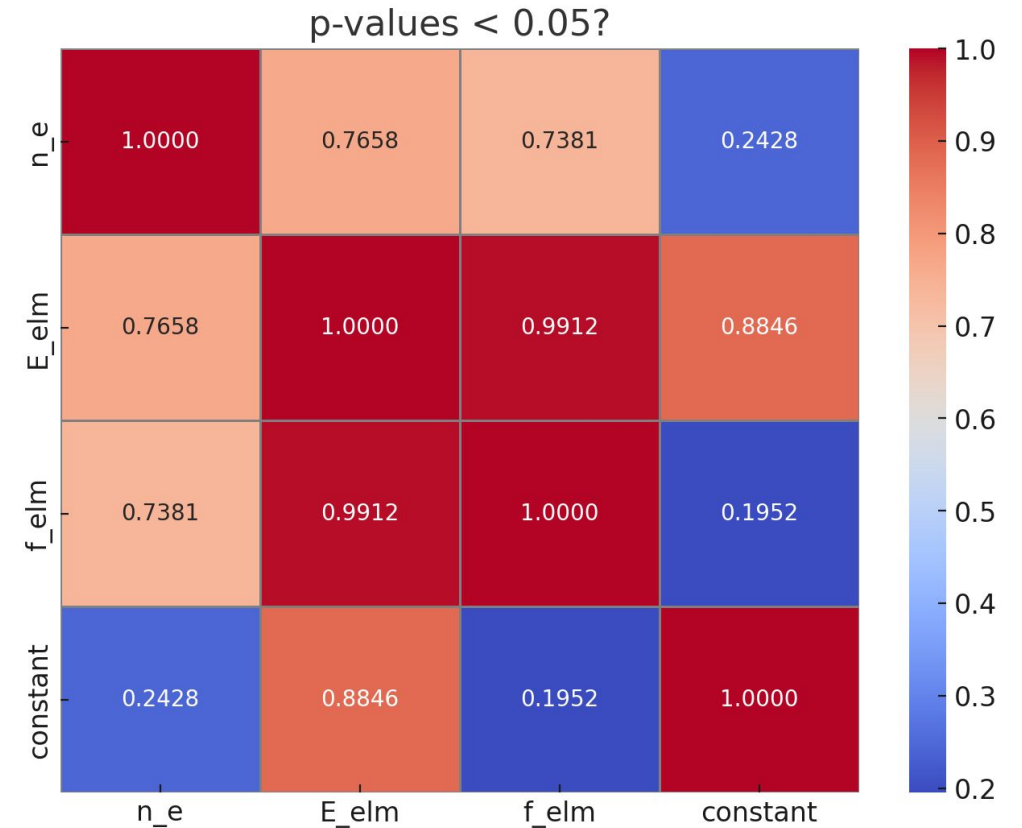
THANK YOU

Hierarchical Bayesian regression



- Dirichlet process
- Sparsifying

R^2 值为: 0.5073

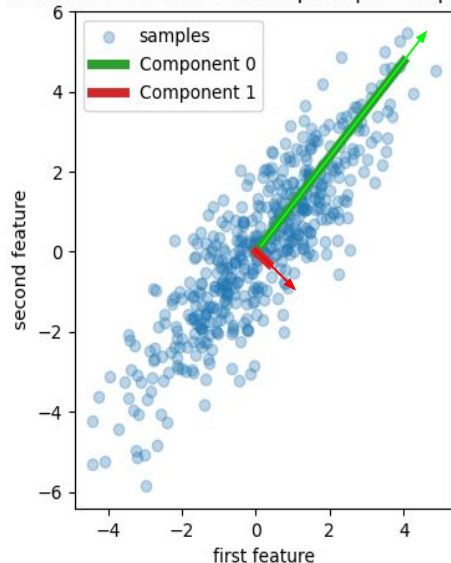


Principal component regression (PCR)

Ridge regression

- L2 regularization

2-dimensional dataset with principal components



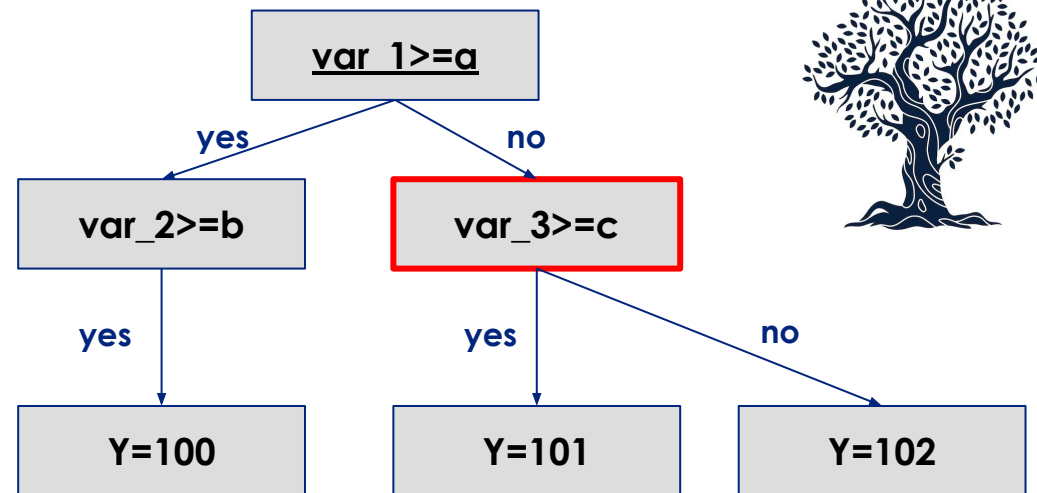
Instability

- Eigenvectors = coordinates
- Eigenvalues = importance

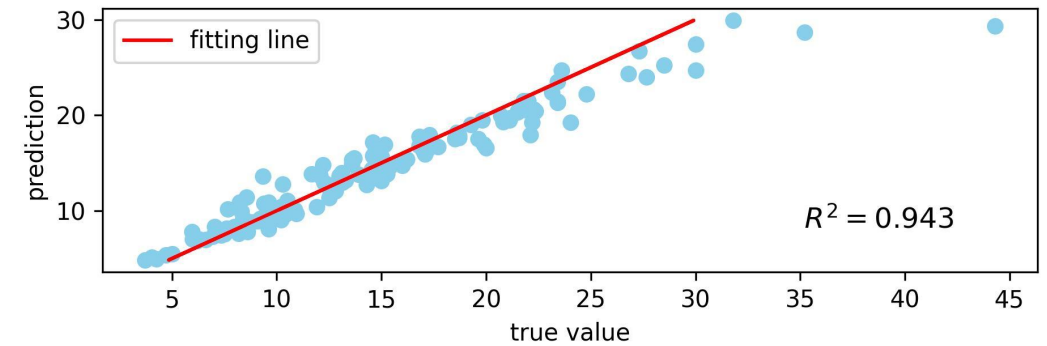
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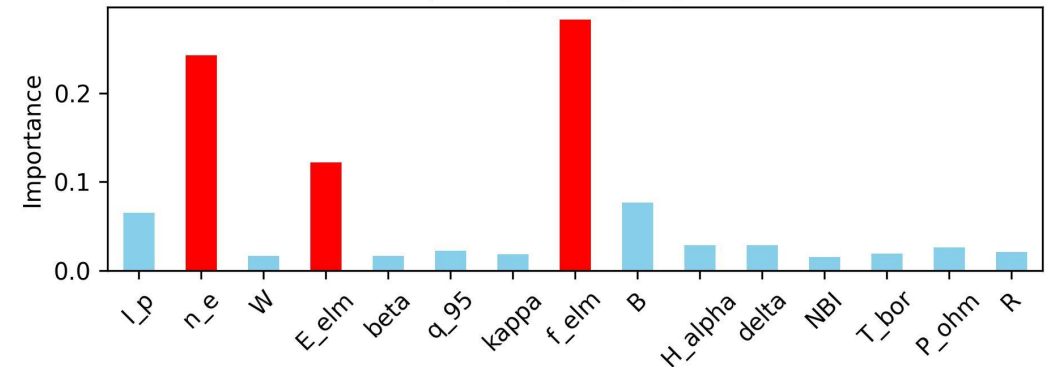
Example: one tree



Regression model



Top 3 signals Highlighted



Importance $\propto$ Number of occurrence